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(54) COMMUNICATION APPARATUS AND CONTROL METHOD

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(57) ABSTRACT

A communication apparatus includes a connection unit that connects with an external apparatus so as to be capable of communication, and a control unit that establishes a plurality of data transfer paths with the external apparatus with which a connection has been established by the connection unit, selects a data transfer path on which transfer is performed, for each image file from among the plurality of data transfer paths, and transfers the image file. The control unit selects a data transfer path on which transfer of a file is possible from among the plurality of data transfer paths, and in a case of transferring an audio file attached to the image file, selects a data transfer path on which transfer of the image file to which the audio file is attached has been performed.

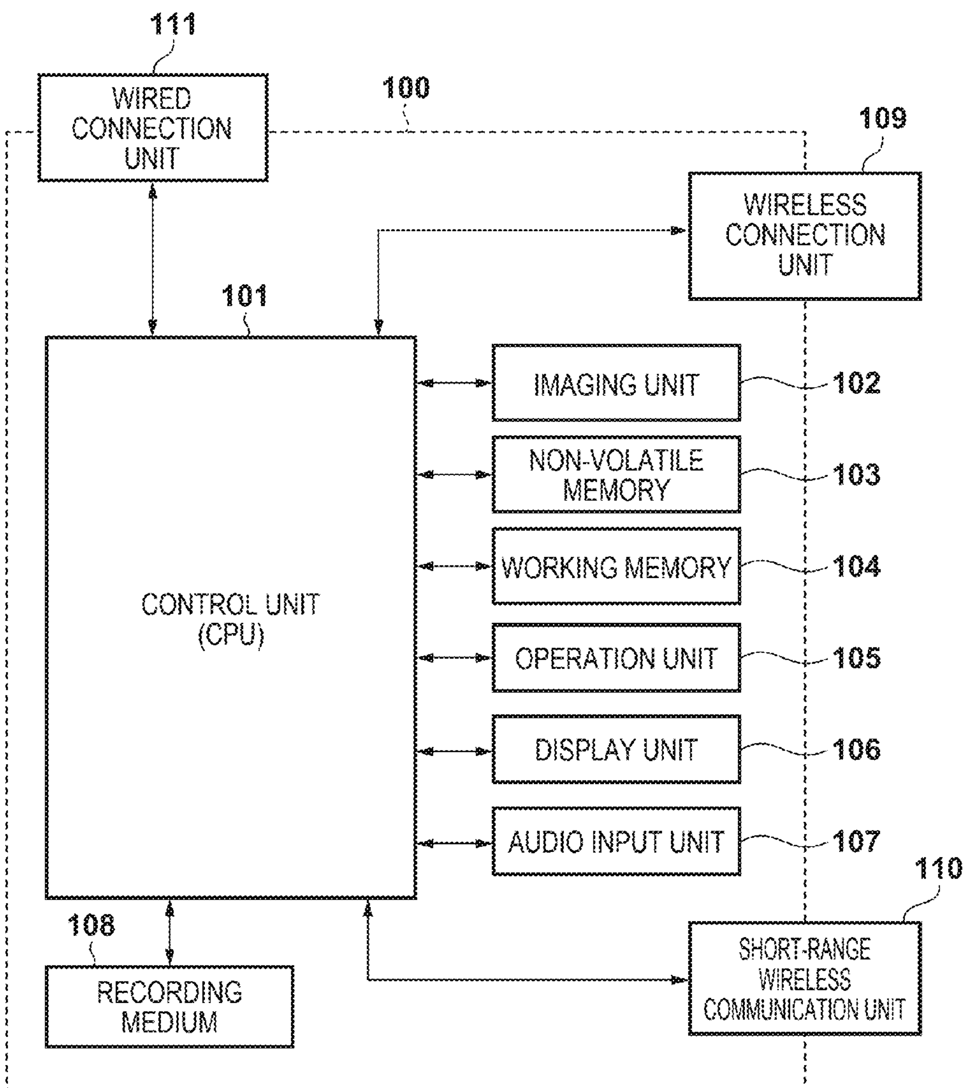
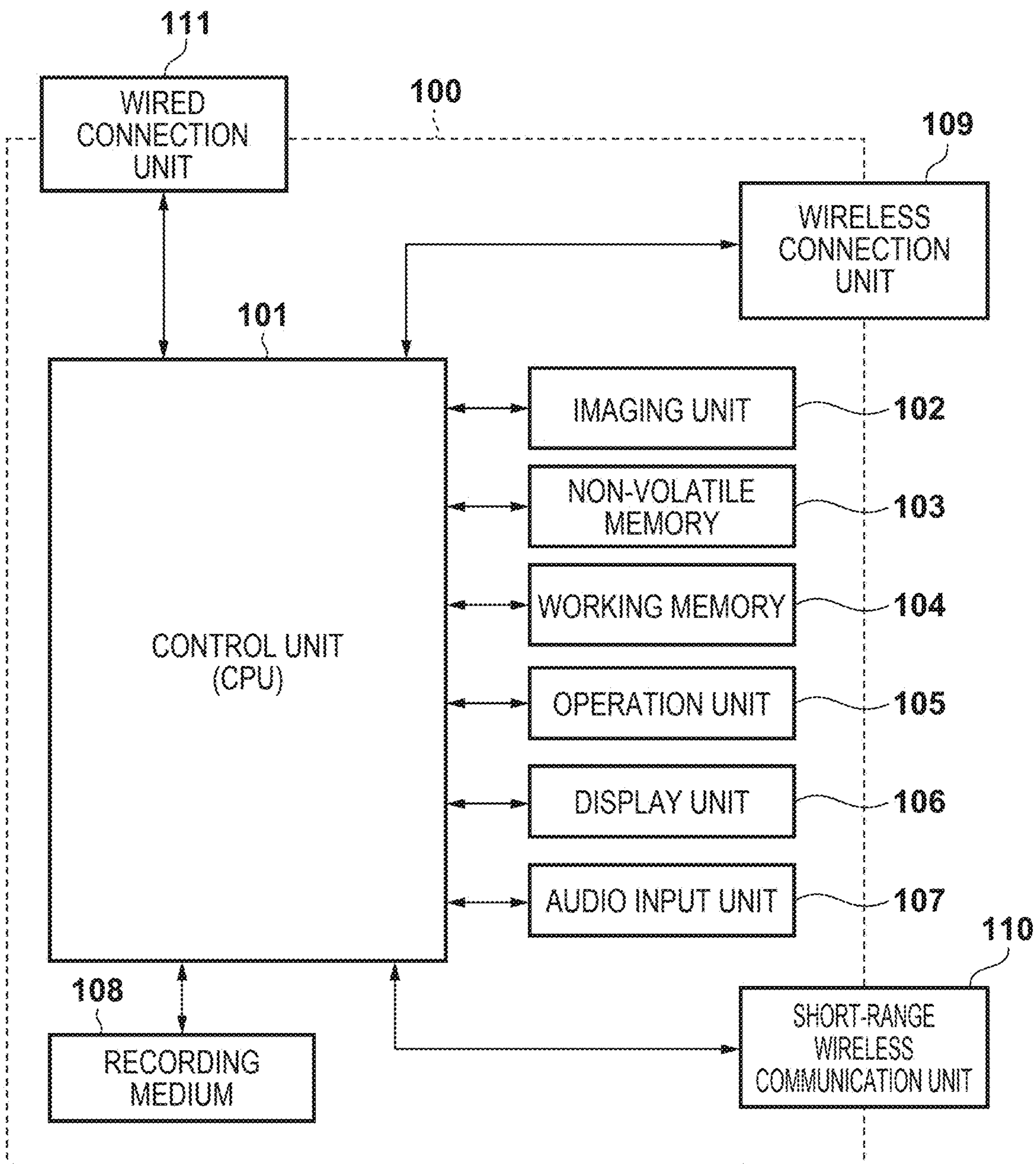


FIG. 1



111
WIRED CONNECTION UNIT
100
109
WIRELESS CONNECTION UNIT
101
CONTROL UNIT (CPU)
IMAGING UNIT
102
NON-VOLATILE MEMORY
103
WORKING MEMORY
104
OPERATION UNIT
105
DISPLAY UNIT
106
AUDIO INPUT UNIT
107
110
108
RECORDING MEDIUM
SHORT-RANGE WIRELESS COMMUNICATION UNIT

FIG. 2A

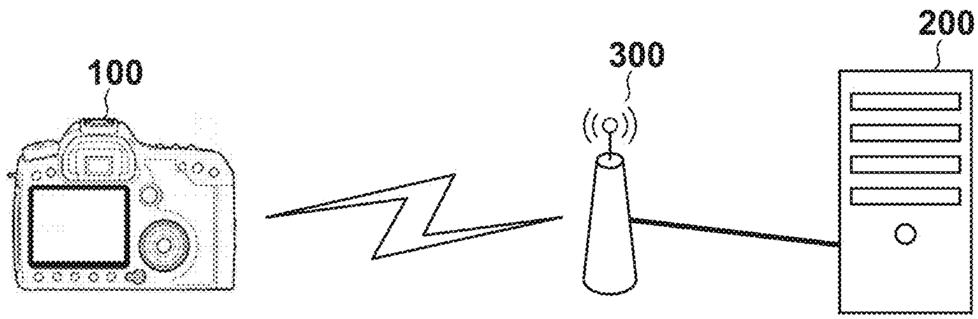


FIG. 2B

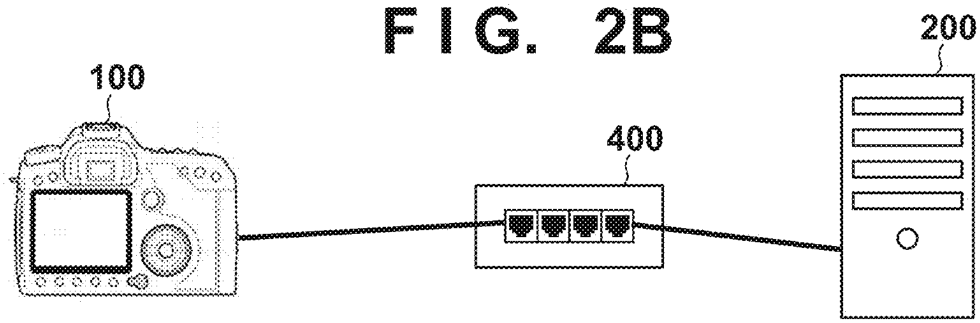


FIG. 3

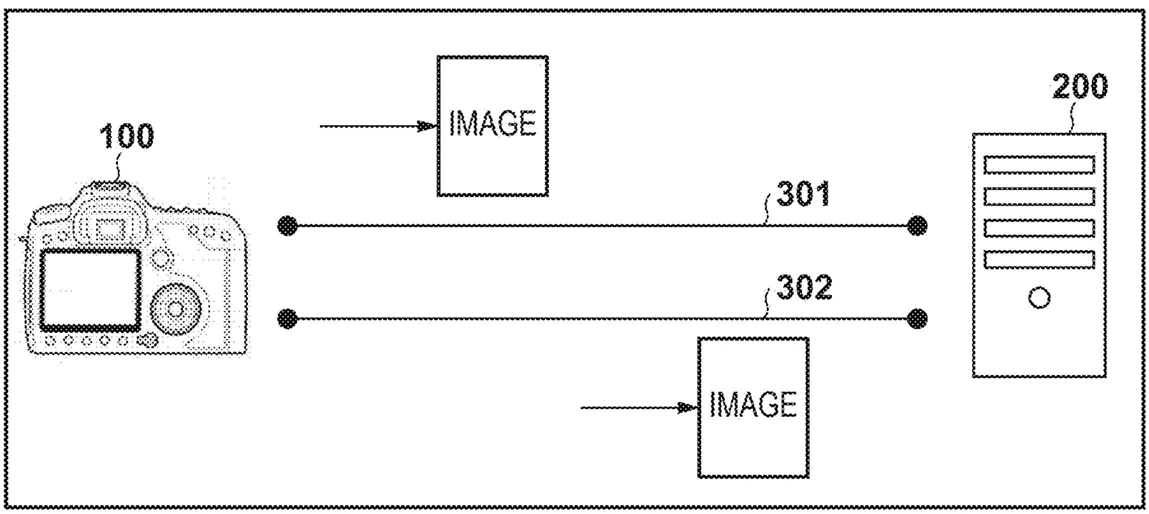


FIG. 4

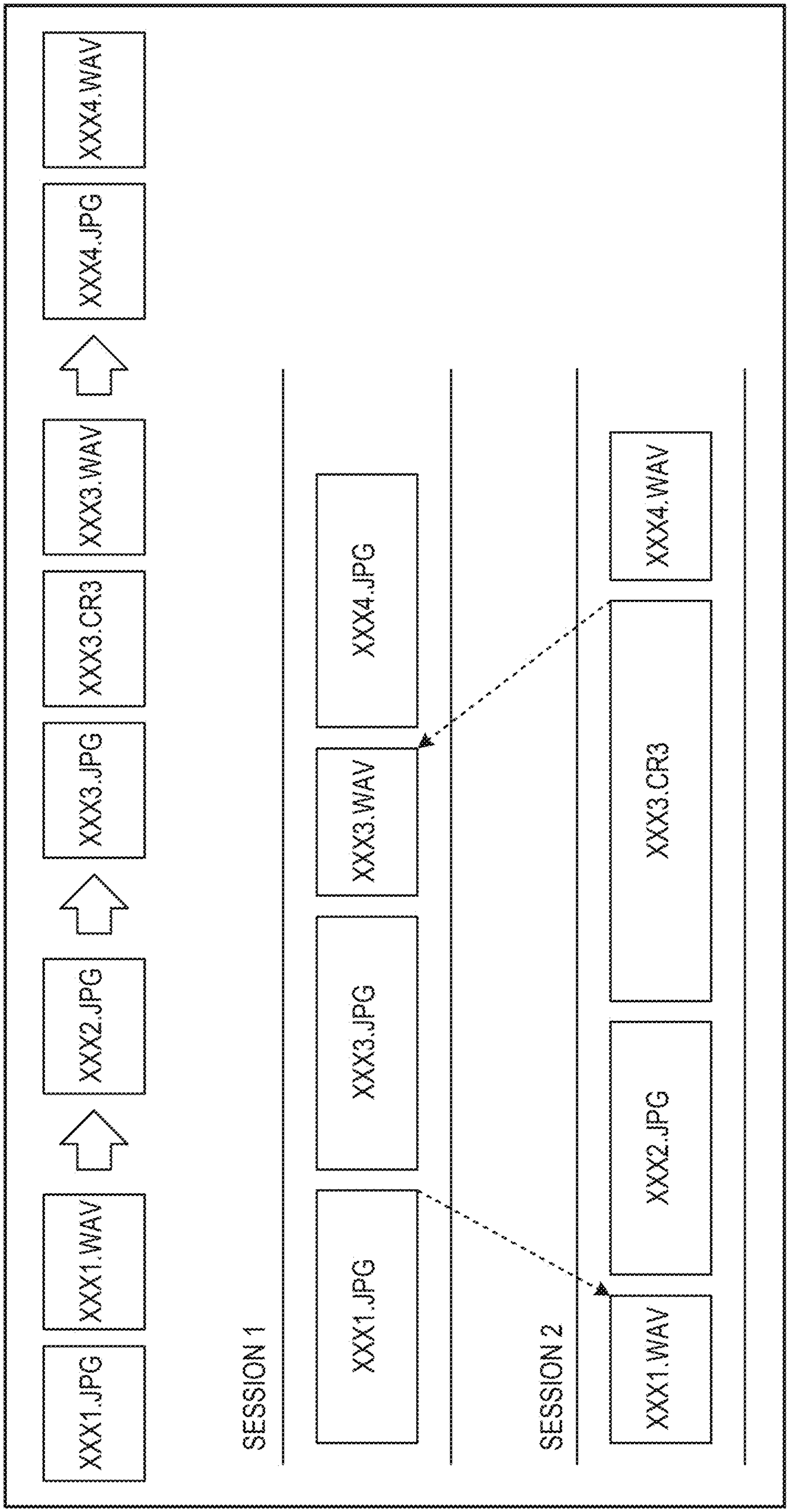

XXX1.JPG
XXX1.WAV
XXX2.JPG
XXX3.JPG
XXX3.CR3
XXX3.WAV
XXX4.JPG
XXX4.WAV
SESSION 1
XXX1.JPG
XXX3.JPG
XXX3.WAV
XXX4.JPG
SESSION 2
XXX1.WAV
XXX2.JPG
XXX3.CR3
XXX4.WAV

FIG. 5A

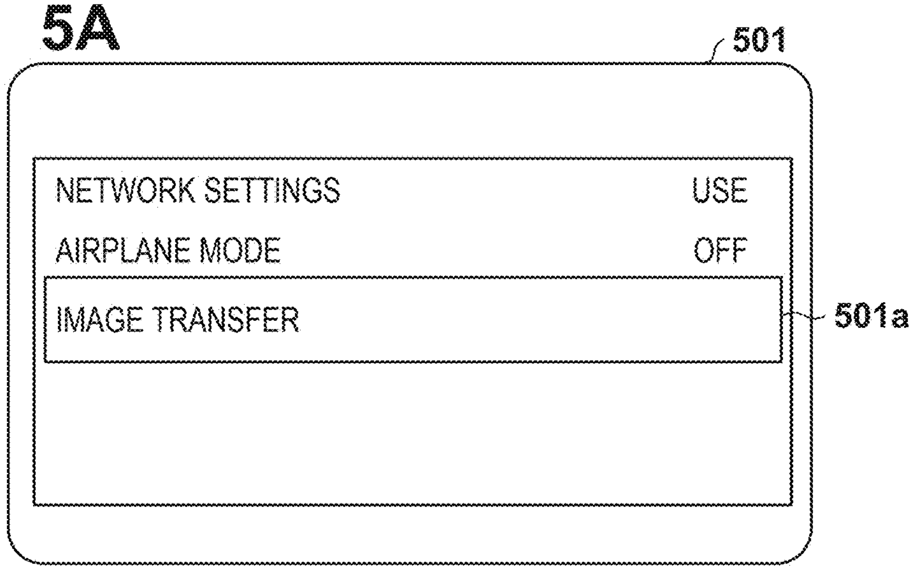


FIG. 5B

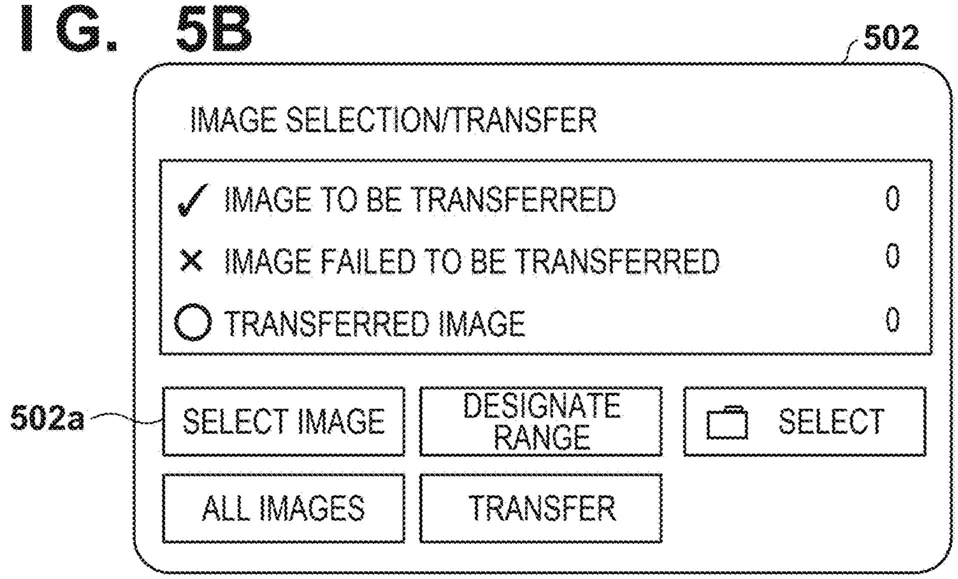


FIG. 5C

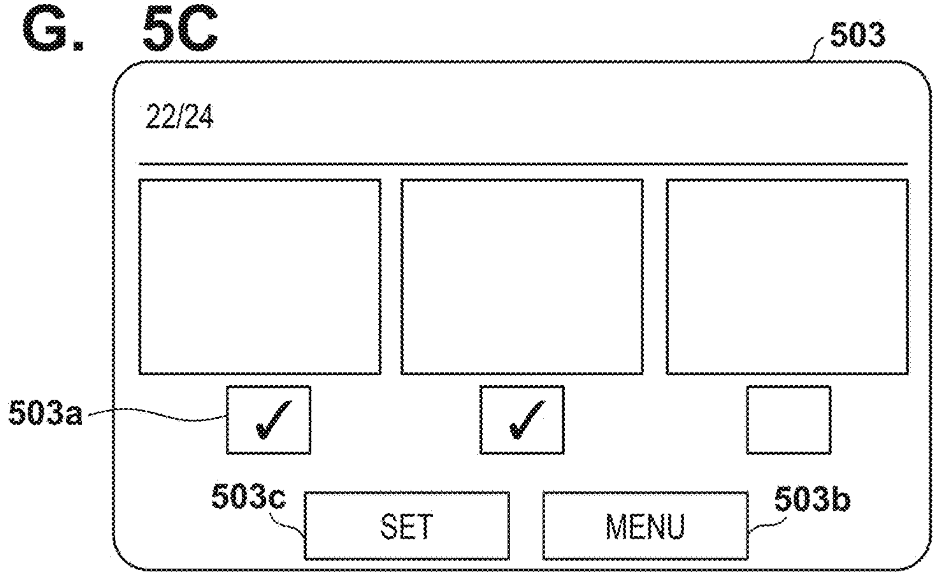


FIG. 5D

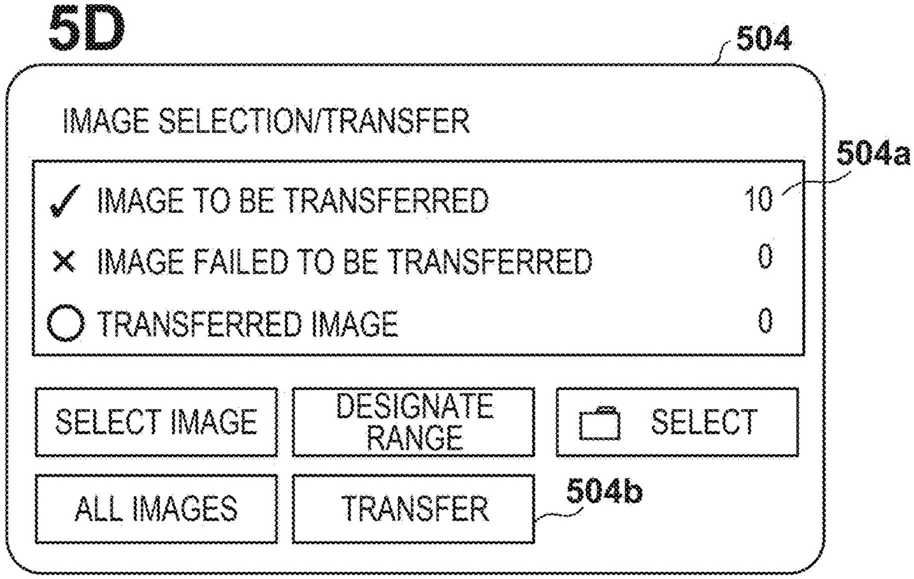


FIG. 5E

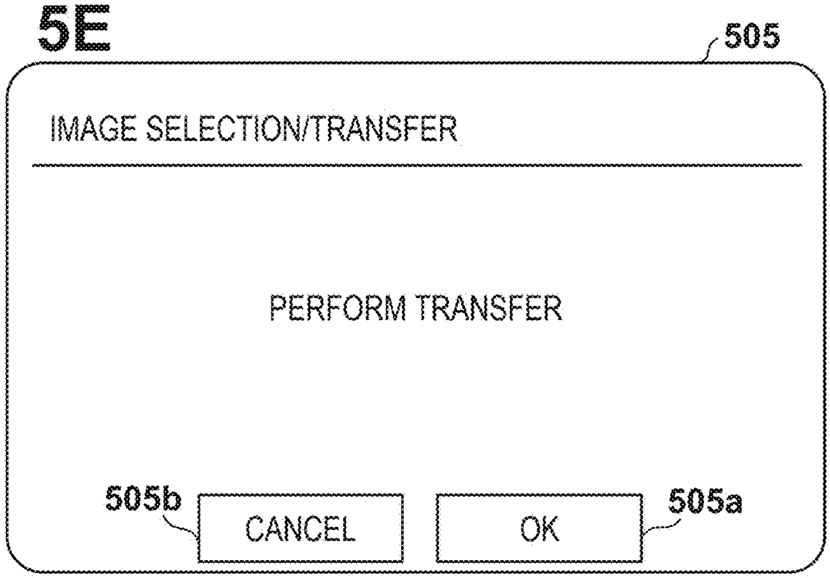


FIG. 5F

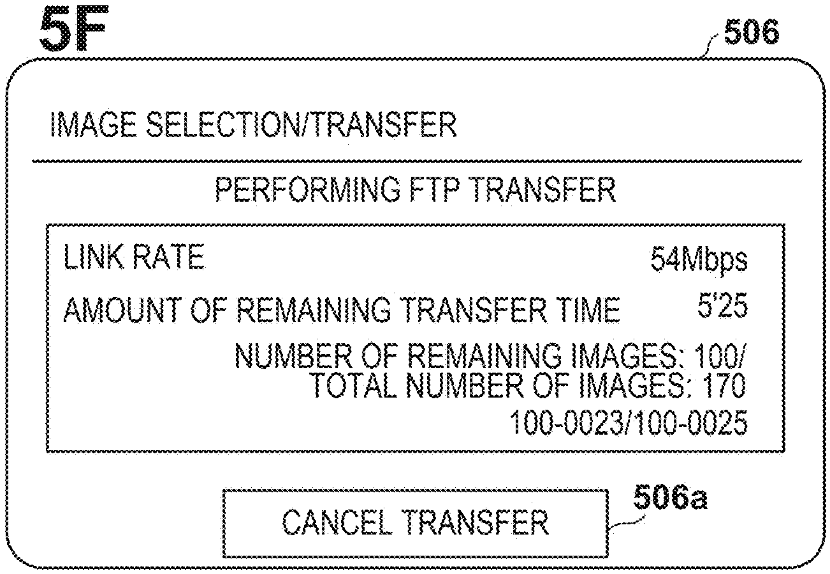
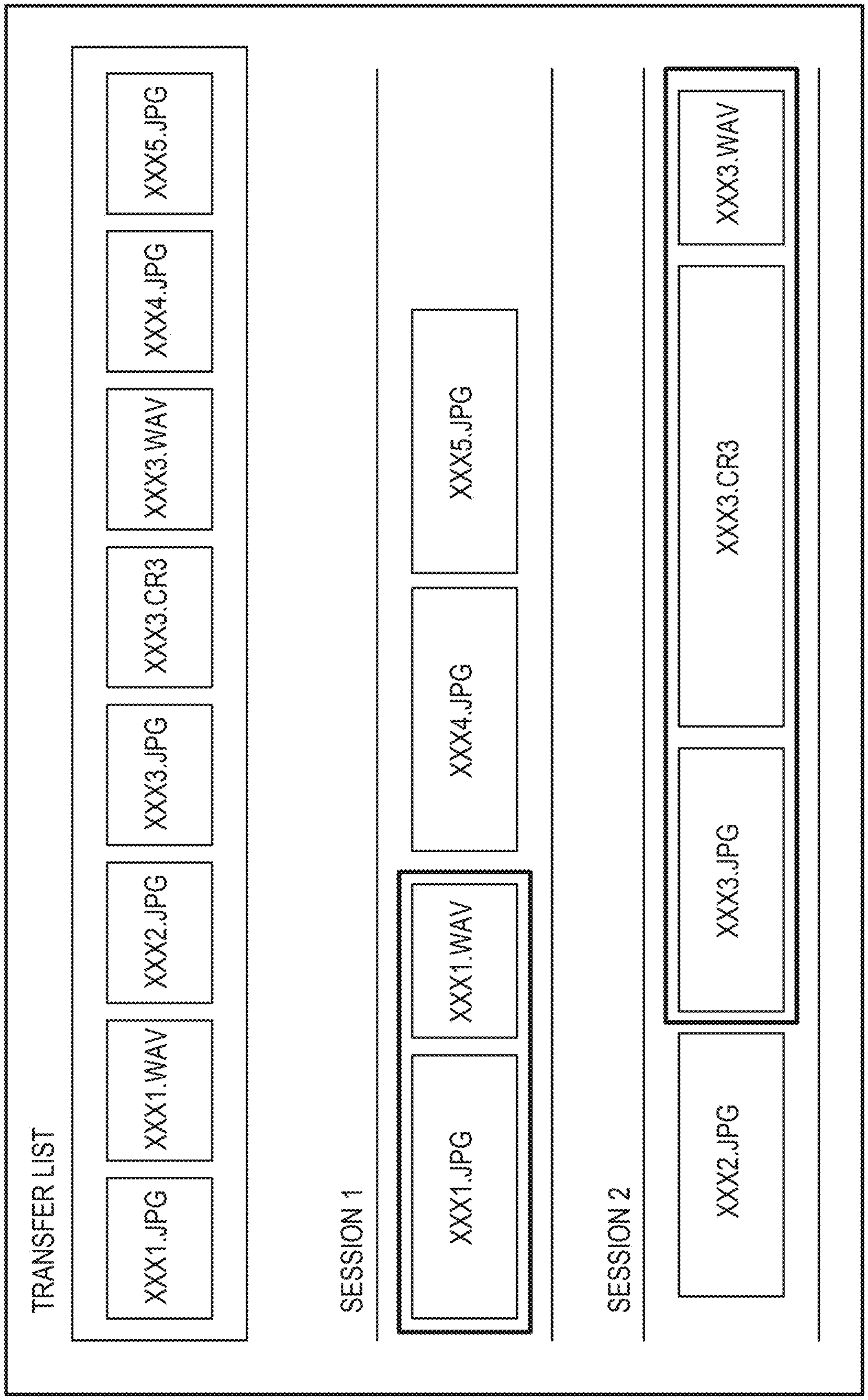


FIG. 6


TRANSFER LIST
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XXX1.WAV
XXX2.JPG
XXX3.JPG
XXX3.CR3
XXX3.WAV
XXX4.JPG
XXX5.JPG
SESSION 1
XXX1.JPG
XXX1.WAV
XXX4.JPG
XXX5.JPG
SESSION 2
XXX2.JPG
XXX3.JPG
XXX3.CR3
XXX3.WAV

FIG. 7A

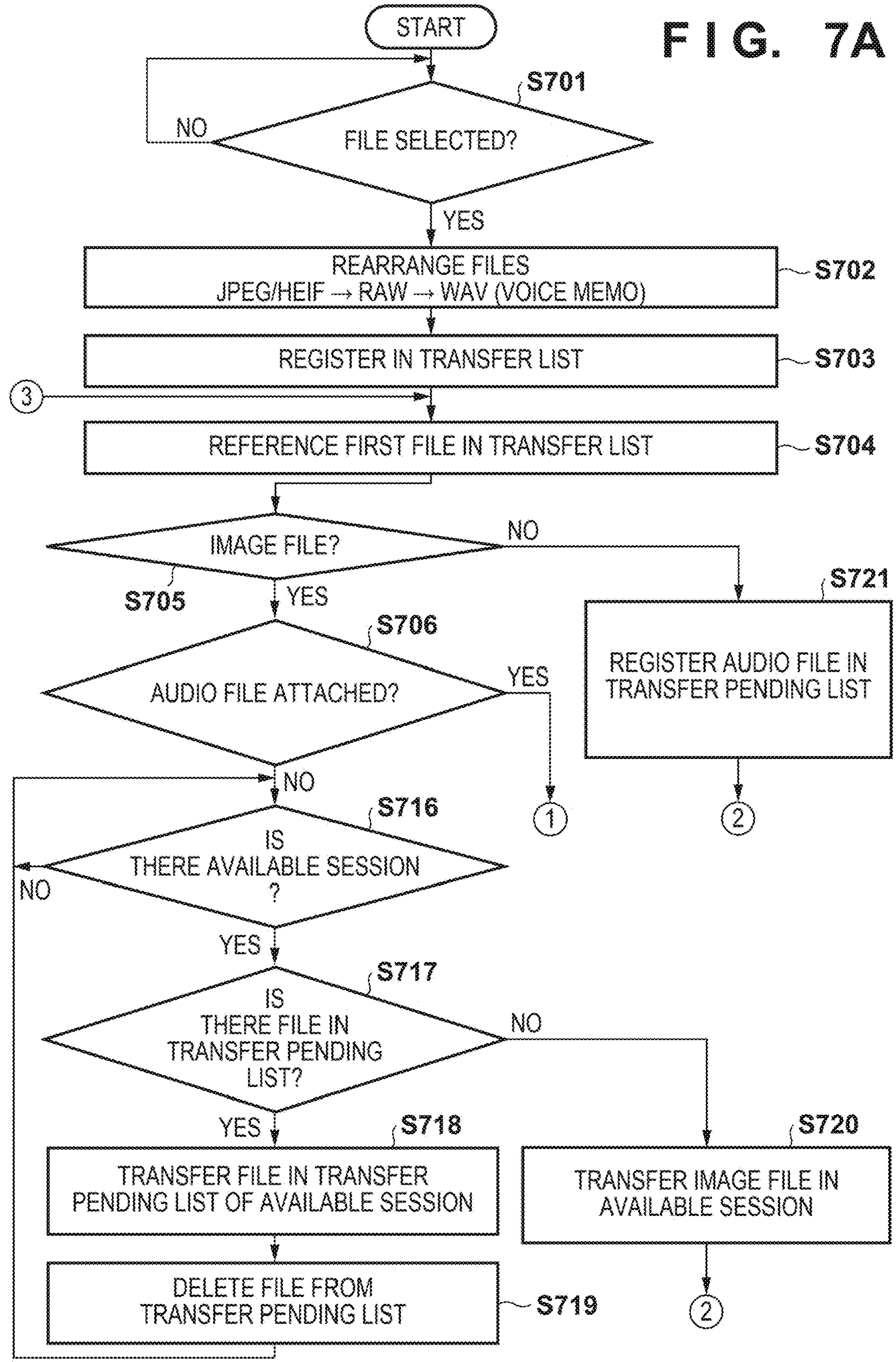

START
S701
FILE SELECTED?
NO
YES
REARRANGE FILES
JPEG/HEIF → RAW → WAV (VOICE MEMO)
S702
REGISTER IN TRANSFER LIST
S703
3
REFERENCE FIRST FILE IN TRANSFER LIST
S704
IMAGE FILE?
S705
NO
YES
S721
REGISTER AUDIO FILE IN TRANSFER PENDING LIST
2
S706
AUDIO FILE ATTACHED?
YES
1
NO
S716
IS THERE AVAILABLE SESSION ?
NO
YES
S717
IS THERE FILE IN TRANSFER PENDING LIST?
NO
YES
S718
TRANSFER FILE IN TRANSFER PENDING LIST OF AVAILABLE SESSION
S720
TRANSFER IMAGE FILE IN AVAILABLE SESSION
2
DELETE FILE FROM TRANSFER PENDING LIST
S719

FIG. 7B

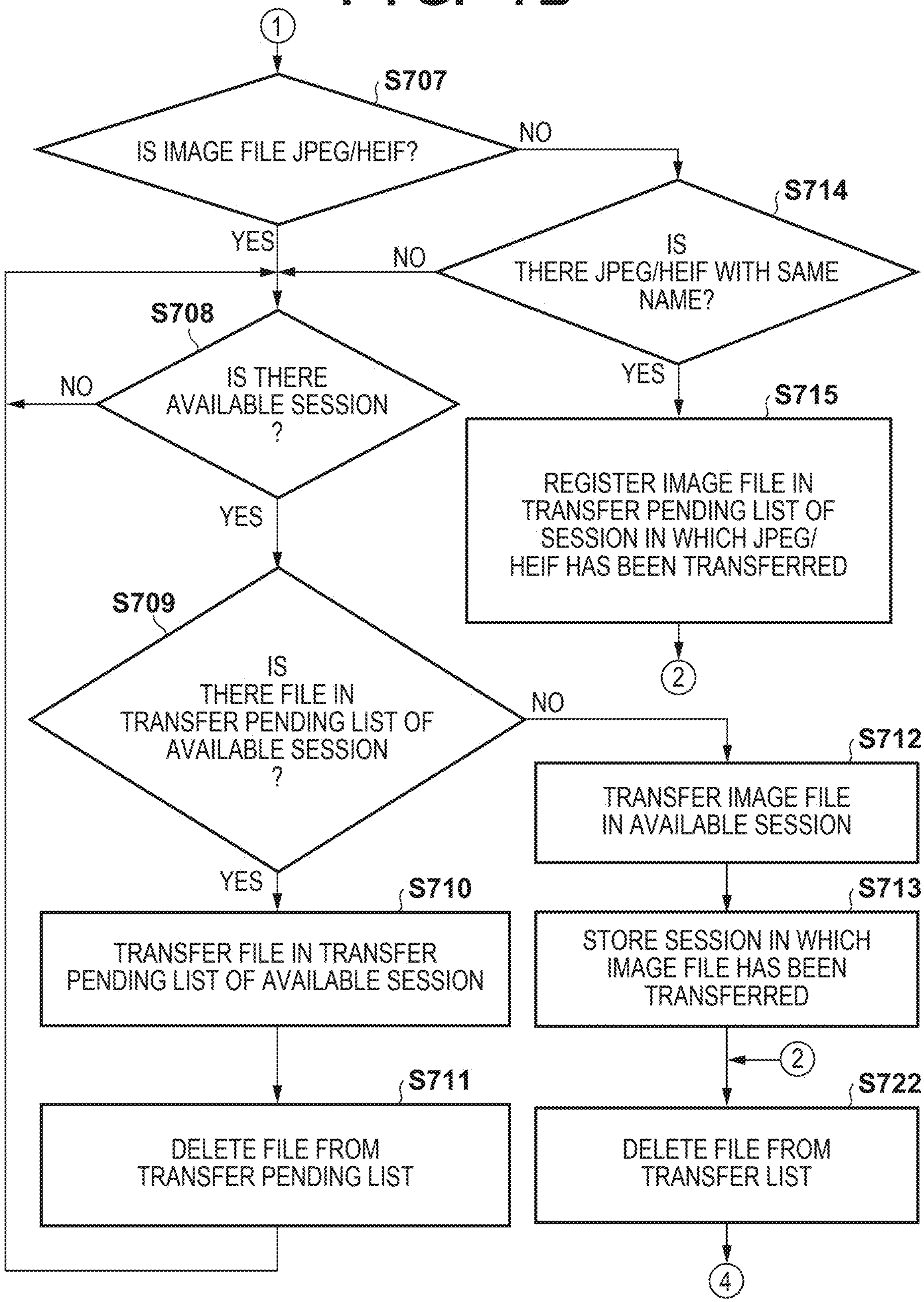

1
S707
IS IMAGE FILE JPEG/HEIF?
NO
YES
S714
IS THERE JPEG/HEIF WITH SAME NAME?
NO
YES
S708
IS THERE AVAILABLE SESSION ?
NO
YES
S715
REGISTER IMAGE FILE IN TRANSFER PENDING LIST OF SESSION IN WHICH JPEG/ HEIF HAS BEEN TRANSFERRED
2
S709
IS THERE FILE IN TRANSFER PENDING LIST OF AVAILABLE SESSION ?
NO
YES
S712
TRANSFER IMAGE FILE IN AVAILABLE SESSION
S713
STORE SESSION IN WHICH IMAGE FILE HAS BEEN TRANSFERRED
S710
TRANSFER FILE IN TRANSFER PENDING LIST OF AVAILABLE SESSION
2
S711
DELETE FILE FROM TRANSFER PENDING LIST
S722
DELETE FILE FROM TRANSFER LIST
4

FIG. 7C

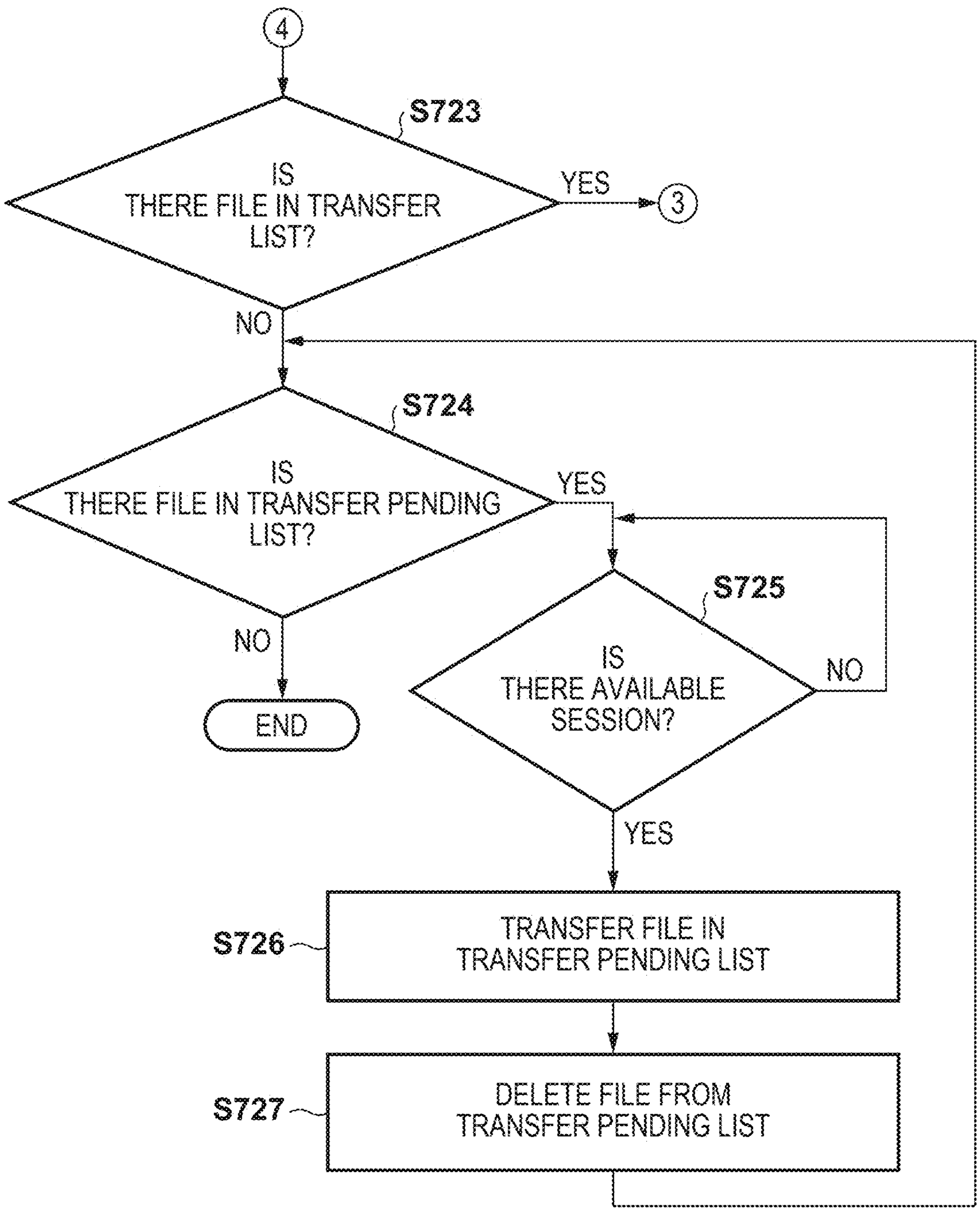
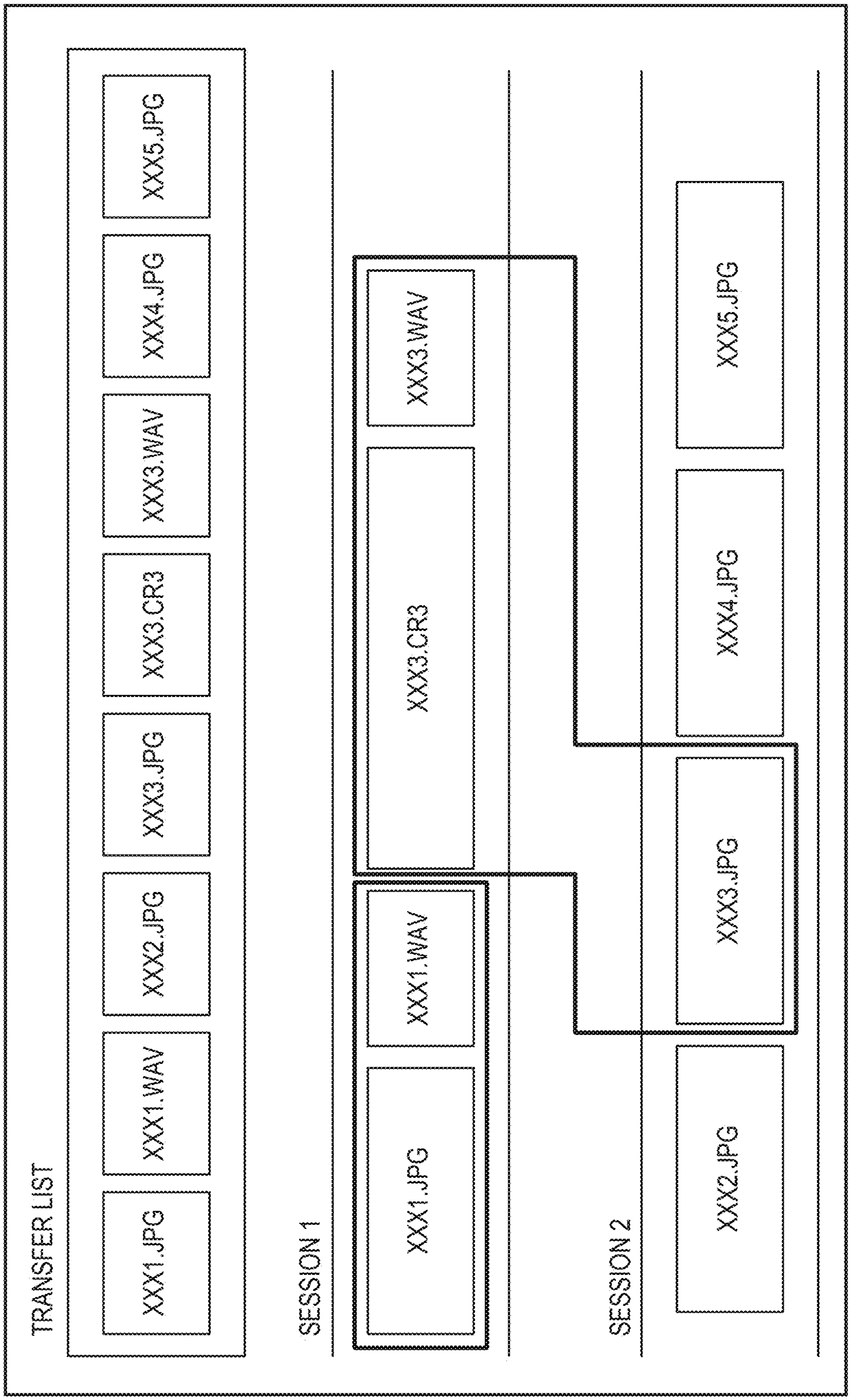

4
S723
IS THERE FILE IN TRANSFER LIST?
YES
3
NO
S724
IS THERE FILE IN TRANSFER PENDING LIST?
YES
NO
END
S725
IS THERE AVAILABLE SESSION?
NO
YES
S726
TRANSFER FILE IN TRANSFER PENDING LIST
S727
DELETE FILE FROM TRANSFER PENDING LIST

FIG. 8


TRANSFER LIST
XXX1.JPG
XXX1.WAV
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XXX3.JPG
XXX3.CR3
XXX3.WAV
XXX4.JPG
XXX5.JPG
SESSION 1
XXX1.JPG
XXX1.WAV
XXX3.CR3
XXX3.WAV
SESSION 2
XXX2.JPG
XXX3.JPG
XXX4.JPG
XXX5.JPG

FIG. 9A

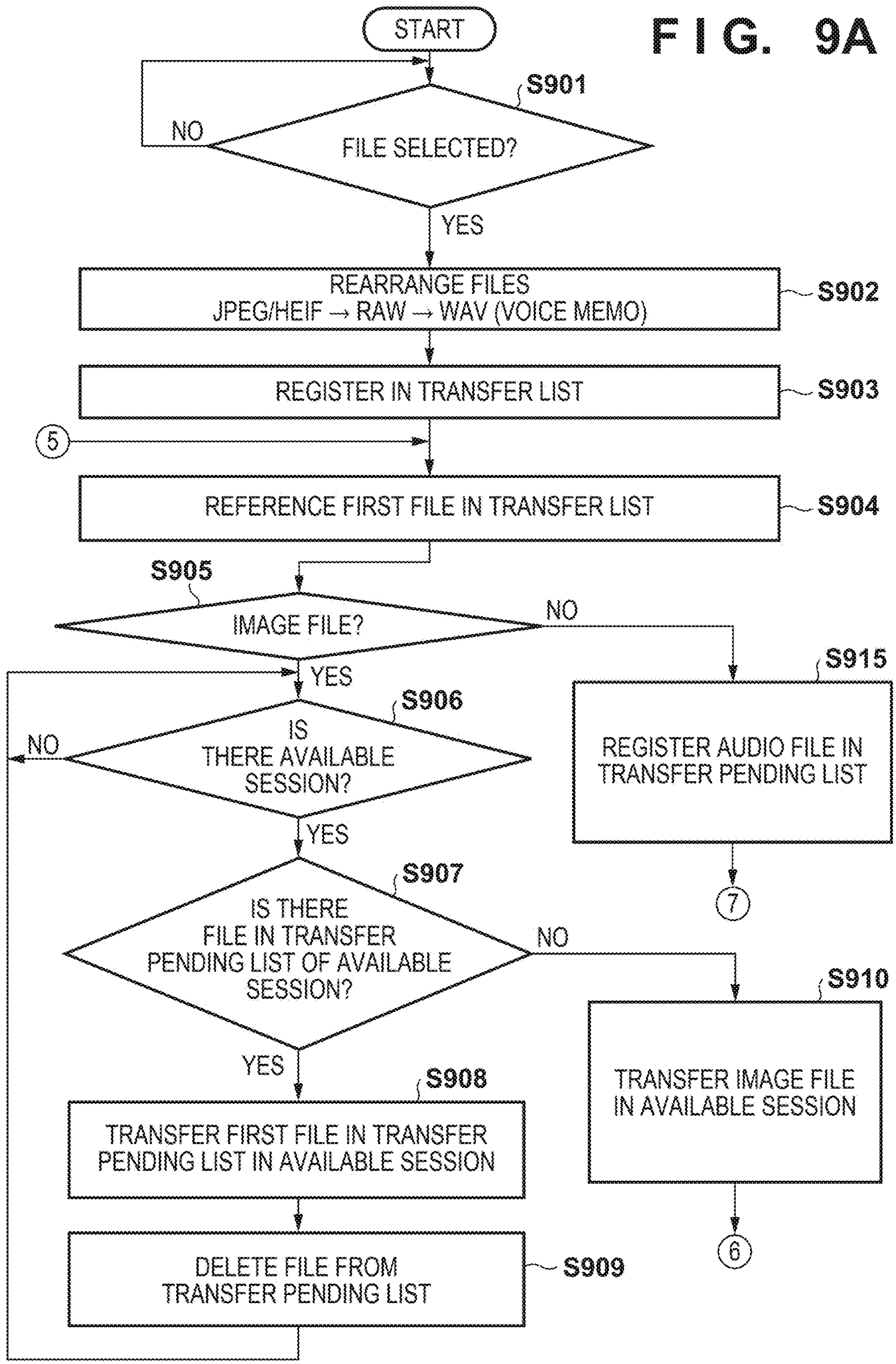
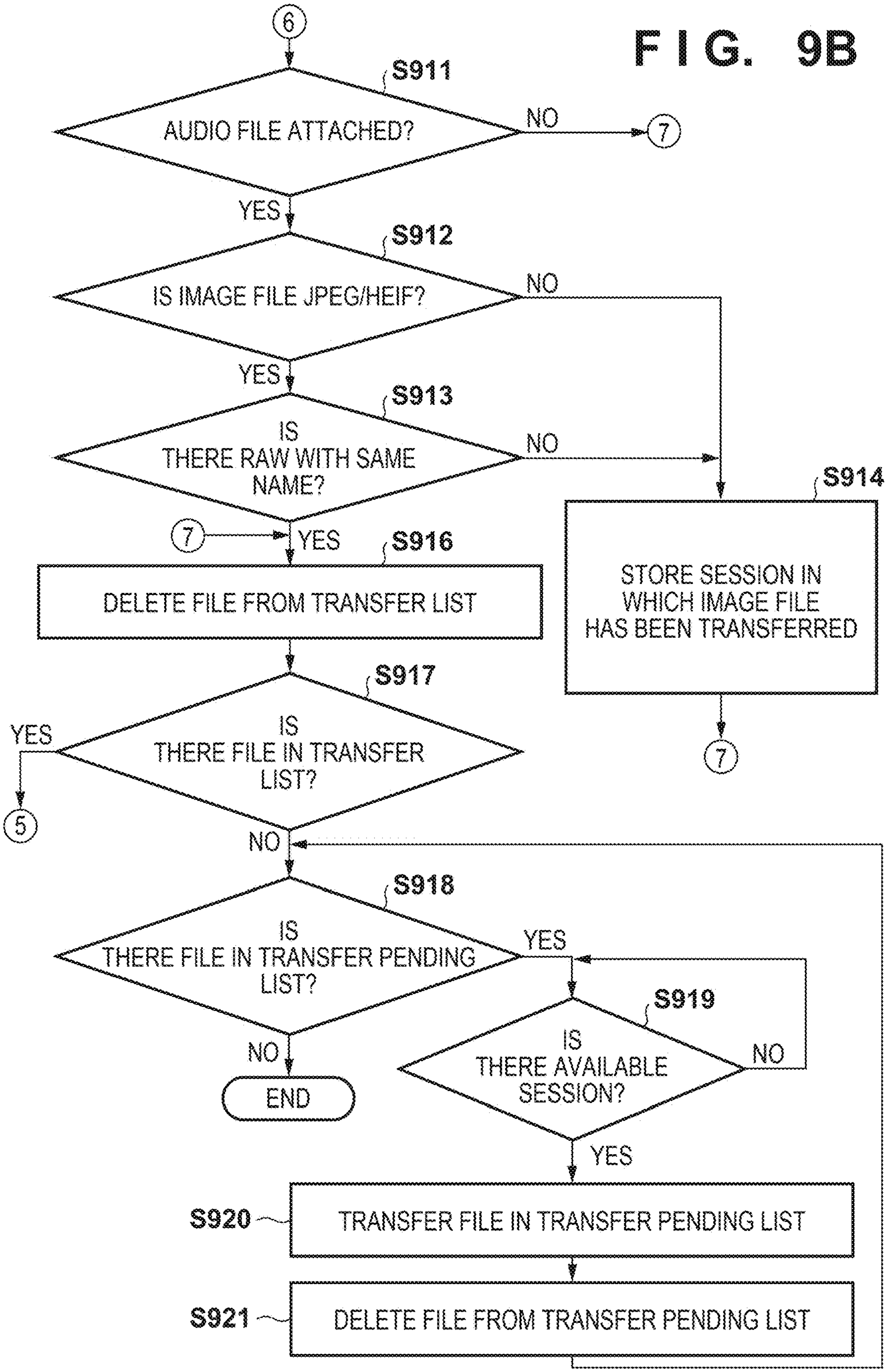
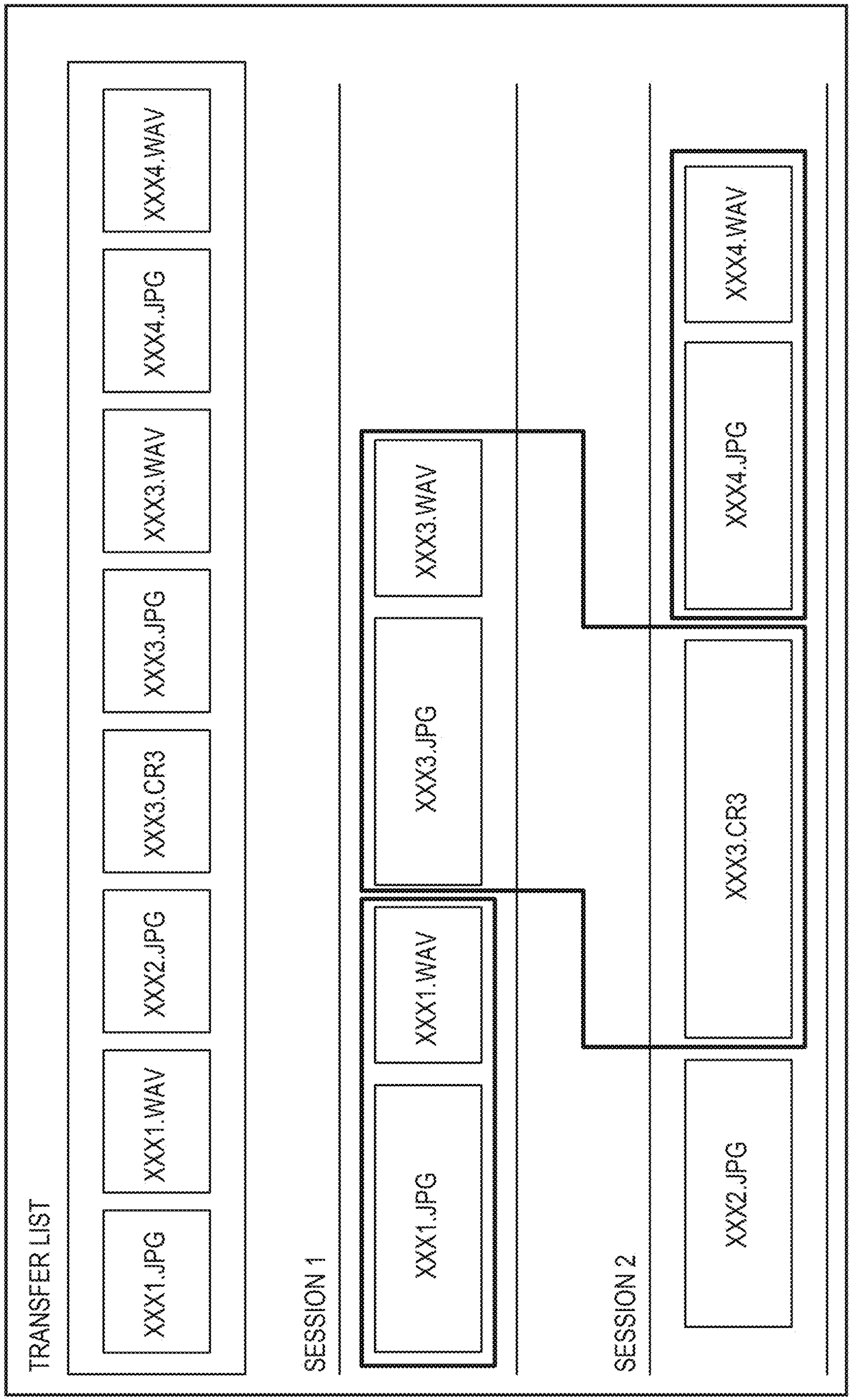

START
S901
FILE SELECTED?
NO
YES
REARRANGE FILES
JPEG/HEIF → RAW → WAV (VOICE MEMO)
S902
REGISTER IN TRANSFER LIST
S903
5
REFERENCE FIRST FILE IN TRANSFER LIST
S904
S905
IMAGE FILE?
NO
YES
S915
REGISTER AUDIO FILE IN TRANSFER PENDING LIST
7
S906
IS THERE AVAILABLE SESSION?
NO
YES
S907
IS THERE FILE IN TRANSFER PENDING LIST OF AVAILABLE SESSION?
NO
YES
S910
TRANSFER IMAGE FILE IN AVAILABLE SESSION
6
S908
TRANSFER FIRST FILE IN TRANSFER PENDING LIST IN AVAILABLE SESSION
DELETE FILE FROM TRANSFER PENDING LIST
S909

FIG. 9B

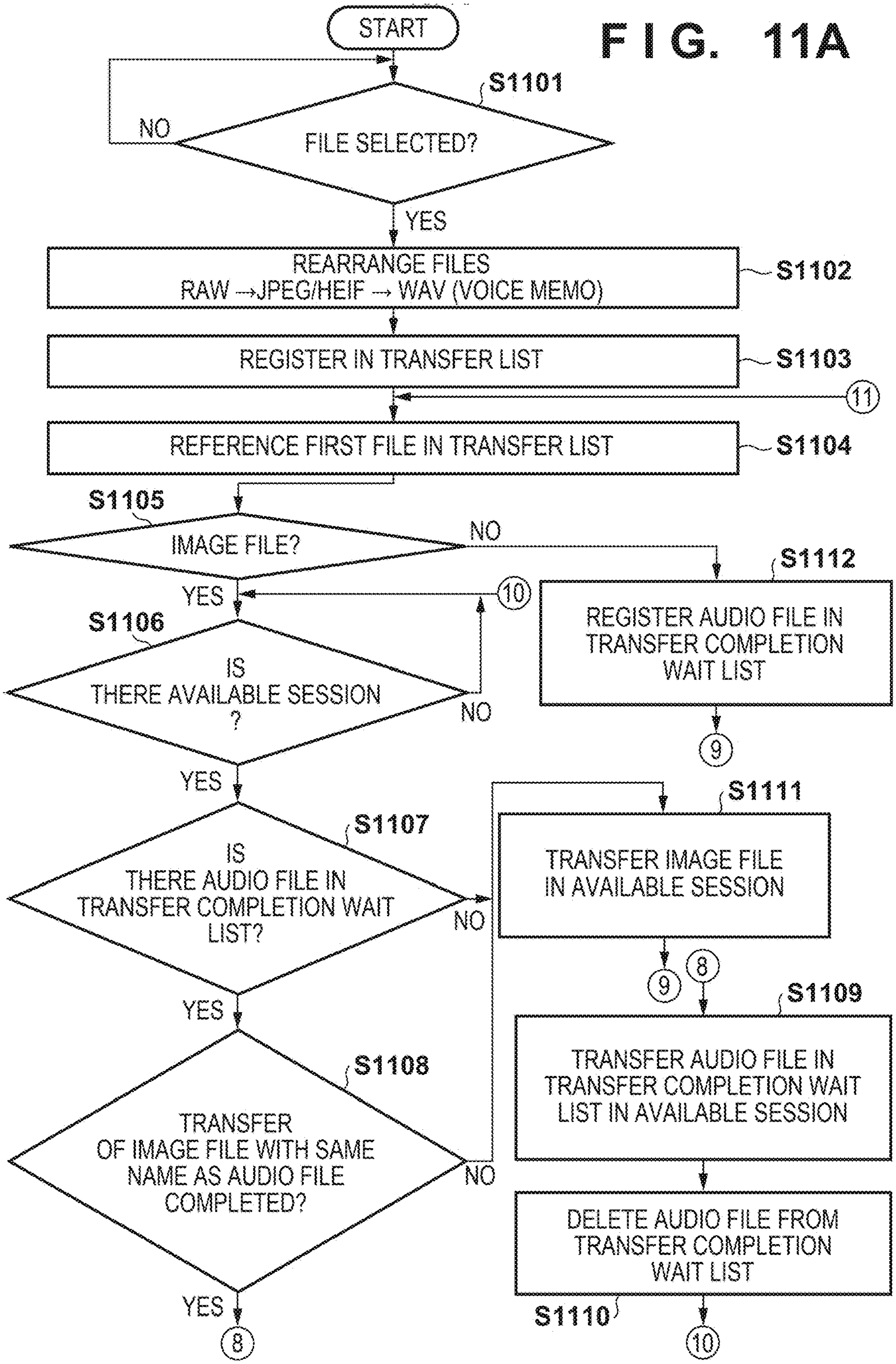


6
S911
AUDIO FILE ATTACHED?
NO
7
YES
S912
IS IMAGE FILE JPEG/HEIF?
NO
YES
S913
IS THERE RAW WITH SAME NAME?
NO
S914
STORE SESSION IN WHICH IMAGE FILE HAS BEEN TRANSFERRED
7
YES
S916
DELETE FILE FROM TRANSFER LIST
7
S917
IS THERE FILE IN TRANSFER LIST?
YES
5
NO
S918
IS THERE FILE IN TRANSFER PENDING LIST?
YES
S919
IS THERE AVAILABLE SESSION?
NO
NO
END
YES
S920
TRANSFER FILE IN TRANSFER PENDING LIST
S921
DELETE FILE FROM TRANSFER PENDING LIST

FIG. 10


TRANSFER LIST
XXX1.JPG
XXX1.WAV
XXX2.JPG
XXX3.CR3
XXX3.JPG
XXX3.WAV
XXX4.JPG
XXX4.WAV
SESSION 1
XXX1.JPG
XXX1.WAV
XXX3.JPG
XXX3.WAV
SESSION 2
XXX2.JPG
XXX3.CR3
XXX4.JPG
XXX4.WAV

F I G. 11A


START
S1101
FILE SELECTED?
NO
YES
REARRANGE FILES
RAW →JPEG/HEIF → WAV (VOICE MEMO)
S1102
REGISTER IN TRANSFER LIST
S1103
11
REFERENCE FIRST FILE IN TRANSFER LIST
S1104
S1105
IMAGE FILE?
NO
YES
10
S1112
REGISTER AUDIO FILE IN TRANSFER COMPLETION WAIT LIST
9
S1106
IS THERE AVAILABLE SESSION ?
NO
YES
S1107
IS THERE AUDIO FILE IN TRANSFER COMPLETION WAIT LIST?
NO
S1111
TRANSFER IMAGE FILE IN AVAILABLE SESSION
9
8
YES
S1108
TRANSFER OF IMAGE FILE WITH SAME NAME AS AUDIO FILE COMPLETED?
NO
S1109
TRANSFER AUDIO FILE IN TRANSFER COMPLETION WAIT LIST IN AVAILABLE SESSION
DELETE AUDIO FILE FROM TRANSFER COMPLETION WAIT LIST
YES
8
S1110
10

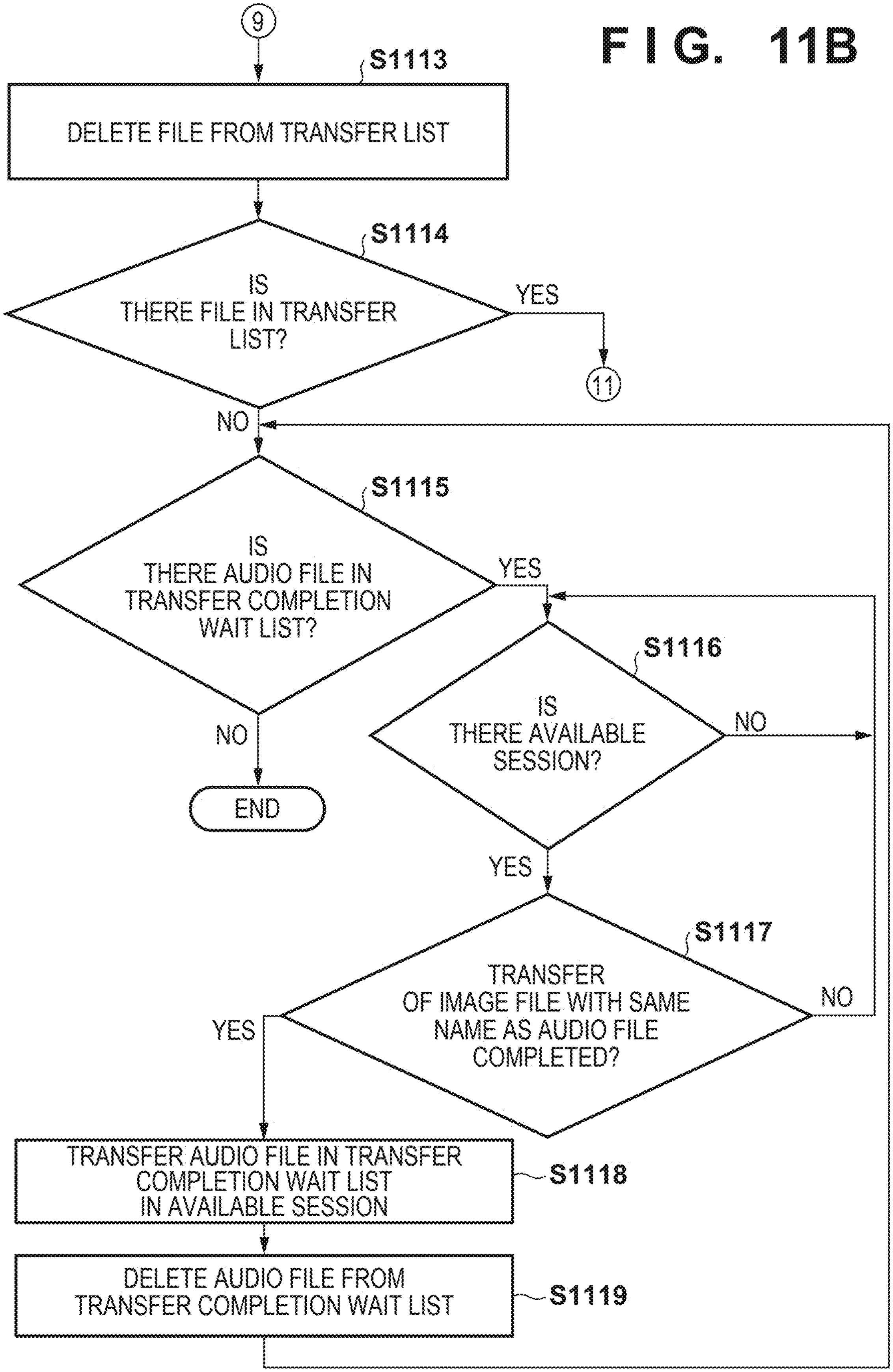

9
S1113
DELETE FILE FROM TRANSFER LIST
S1114
IS THERE FILE IN TRANSFER LIST?
YES
11
NO
S1115
IS THERE AUDIO FILE IN TRANSFER COMPLETION WAIT LIST?
YES
NO
END
S1116
IS THERE AVAILABLE SESSION?
NO
YES
S1117
TRANSFER OF IMAGE FILE WITH SAME NAME AS AUDIO FILE COMPLETED?
YES
NO
TRANSFER AUDIO FILE IN TRANSFER COMPLETION WAIT LIST IN AVAILABLE SESSION
S1118
DELETE AUDIO FILE FROM TRANSFER COMPLETION WAIT LIST
S1119

FIG. 11B

COMMUNICATION APPARATUS AND CONTROL METHOD

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a Continuation of International Patent Application No. PCT/JP2023/031638, filed Aug. 30, 2023, which claims the benefits of Japanese Patent Application No. 2022-193607, filed Dec. 2, 2022, both of which are hereby incorporated by reference herein in their entirety.

BACKGROUND

Field of the Technology

[0002] The present disclosure relates to a technique for transferring files from a communication apparatus to an external apparatus.

Description of the Related Art

[0003] PTL 1 describes a digital camera for selecting either a server, a personal computer (PC), or a printer and transferring an image file. Some digital cameras include a function for recording, as audio, image shooting conditions, image description, and the like and attaching the audio file as a voice memo to an image file. The voice memo can be transferred with the image file to an external apparatus, and reproduction of the voice memo and editing and the like of the image file can be performed in the external apparatus.

CITATION LIST

Patent Literature

[0004] PTL1: Japanese Patent Laid-Open No. 2011-120279

[0005] When editing and the like of an image file to which a voice memo is attached are performed with reception of a voice memo by an external apparatus as a trigger, it is necessary that transfer be performed such that the voice memo arrives at the external apparatus after the image file. Further, when the external apparatus is an FTP server, throughput can be improved by transferring a plurality of image files in a plurality of FTP sessions, but voice memos may end up arriving at the FTP server before the image files.

SUMMARY

[0006] The present disclosure has been made in consideration of the aforementioned problems and realizes techniques for performing control such that a voice memo arrives later than an image file when transferring the image file to which the voice memo is attached to an external apparatus.

[0007] In order to solve the aforementioned problems, the present disclosure provides a communication apparatus comprising: a connection unit that connects with an external apparatus so as to be capable of communication; and a control unit that establishes a plurality of data transfer paths with the external apparatus with which a connection has been established by the connection unit, selects a data transfer path on which transfer is performed, for each image file from among the plurality of data transfer paths, and transfers the image file, wherein the control unit selects a data transfer path on which transfer of a file is possible from among the plurality of data transfer paths, and in a case of transferring an audio file attached to the image file, selects a data transfer path on which transfer of the image file to which the audio file is attached has been performed.

[0008] In order to solve the aforementioned problems, the present disclosure provides a control method of a communication apparatus having a connection unit that connects with an external apparatus so as to be capable of communication, the method comprising: performing control so as to establish a plurality of data transfer paths with the external apparatus with which a connection has been established by the connection unit, select a data transfer path on which transfer is performed, for each image file from among the plurality of data transfer paths, and transfer the image file, wherein in the control, a data transfer path on which transfer of a file is possible is selected from among the plurality of data transfer paths, and in a case of transferring an audio file attached to the image file, a data transfer path on which transfer of the image file to which the audio file is attached has been performed is selected.

[0009] Features of the present disclosure will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] FIG. **1** is a block diagram illustrating a configuration of a digital camera according to Embodiment 1.

[0011] FIG. **2**A is a diagram illustrating a form of connection between the digital camera and an FTP server according to Embodiment 1.

[0012] FIG. **2**B is a diagram illustrating a form of connection between the digital camera and the FTP server according to Embodiment 1.

[0013] FIG. **3** is a diagram illustrating a method of transferring image files from the digital camera to the FTP server according to Embodiment 1.

[0014] FIG. **4** is a diagram illustrating a conventional problem that occurs when transferring an image file and a voice memo from the digital camera to the FTP server.

[0015] FIG. **5**A is a diagram illustrating a UI screen for selecting an image file to be transferred from the digital camera to the FTP server according to Embodiment 1.

[0016] FIG. **5**B is a diagram illustrating a UI screen for selecting an image file to be transferred from the digital camera to the FTP server according to Embodiment 1.

[0017] FIG. **5**C is a diagram illustrating a UI screen for selecting an image file to be transferred from the digital camera to the FTP server according to Embodiment 1.

[0018] FIG. **5**D is a diagram illustrating a UI screen for selecting an image file to be transferred from the digital camera to the FTP server according to Embodiment 1.

[0019] FIG. **5**E is a diagram illustrating a UI screen for selecting an image file to be transferred from the digital camera to the FTP server according to Embodiment 1.

[0020] FIG. **5**F is a diagram illustrating a UI screen for selecting an image file to be transferred from the digital camera to the FTP server according to Embodiment 1.

[0021] FIG. **6** is a diagram illustrating a method of transferring an image file and a voice memo from the digital camera to the FTP server according to Embodiment 1.

[0022] FIG. **7**A is a flowchart showing control processing for transferring an image file and a voice memo from the digital camera to the FTP server according to Embodiment 1.

[0023] FIG. **7**B is a flowchart showing control processing for transferring an image file and a voice memo from the digital camera to the FTP server according to Embodiment 1.

[0024] FIG. **7**C is a flowchart showing control processing for transferring an image file and a voice memo from the digital camera to the FTP server according to Embodiment 1.

[0025] FIG. **8** is a diagram illustrating a method of transferring an image file and a voice memo from the digital camera to the FTP server according to Embodiment 2.

[0026] FIG. **9**A is a flowchart showing control processing for transferring an image file and a voice memo from the digital camera to the FTP server according to Embodiment 2.

[0027] FIG. **9**B is a flowchart showing control processing for transferring an image file and a voice memo from the digital camera to the FTP server according to Embodiment 2.

[0028] FIG. **10** is a diagram illustrating a method of transferring an image file and a voice memo from the digital camera to the FTP server according to Embodiment 3.

[0029] FIG. **11**A is a flowchart showing control processing for transferring an image file and a voice memo from the digital camera to the FTP server according to Embodiment 3.

[0030] FIG. **11**B is a flowchart showing control processing for transferring an image file and a voice memo from the digital camera to the FTP server according to Embodiment 3.

DESCRIPTION OF THE EMBODIMENTS

[0031] Hereinafter, embodiments will be described in detail with reference to the attached drawings. Note, the following embodiments are not intended to limit the scope of the claimed disclosure. Multiple features are described in the embodiments, but limitation is not made a disclosure that requires all such features, and multiple such features may be combined as appropriate. Furthermore, in the attached drawings, the same reference numerals are given to the same or similar configurations, and redundant description thereof is omitted.

[0032] An example in which a communication apparatus according to the present disclosure has been applied to a digital camera capable of shooting still images and moving images will be described below. In the present embodiment, an example in which the communication apparatus according to the present disclosure has been applied to a digital camera will be described, but the present disclosure is not limited to this example and can be applied to any apparatus that includes a camera function and a communication function. That is, the present disclosure is applicable to mobile devices (e.g., personal computers (desktop PCs and notebook PCs) tablet PCs, smartphones, smartwatches, and smart glasses) and application-specific devices (e.g., music players, game consoles, and electronic book readers) that include a camera function.

[0033] A digital camera according to the present embodiment includes a voice memo function for generating an audio file in which image shooting conditions, image description, and the like are recorded and attaching the audio file to an image file. The audio file generated by the voice memo function can be transferred with the image file to an external apparatus, and reproduction of the audio file and editing and the like of the image file can be performed in the external apparatus. An external apparatus according to the present embodiment is an FTP server that transmits and receives files to and from an FTP client according to File Transfer Protocol (FTP). The digital camera according to the present embodiment is connected with the FTP server so as to be capable of communication as an FTP client. A communication protocol according to the present embodiment is not limited to FTP and may be File Transfer Protocol over SSL/TLS (FTPS), SSH File Transfer Protocol (SFTP), and the like.

[0034] In the present embodiment, processing for establishing a plurality of FTP sessions (control connections and data connections) as a plurality of data transfer paths between the digital camera and the FTP server and performing control such that an audio file, which has been generated by the voice memo function, arrives at the FTP server later than an image file when transferring the image file and the audio file from the digital camera to the FTP server in the plurality of FTP sessions will be described.

Embodiment 1

[0035] First, Embodiment 1 will be described with reference to FIGS. **1** to **7**C.

<Configuration of Digital Camera **100**>

[0036] First, a configuration and functions of a digital camera **100** according to the present embodiment will be described with reference to FIG. **1**.

[0037] A control unit **101** is a central processing unit (CPU) that comprehensively controls the entire digital camera **100** and realizes communication processing and control processing, which will be described below, by executing a program stored in a non-volatile memory **103**, which will be described below. Instead of being controlled by the control unit **101**, the entire apparatus may be controlled by a plurality of pieces of hardware sharing processing.

[0038] An imaging unit **102** includes a group of lenses that include a zoom lens and a focus lens and a shutter that includes an aperture function. Further, the imaging unit **102** includes an imaging element constituted by a CCD, a CMOS device, or the like for converting a subject image into an electrical signal and an A/D converter for converting an analog image signal outputted from the imaging element into a digital signal. Under the control of the control unit **101**, the imaging unit **102** converts subject image light imaged by the lenses included in the imaging unit **102** into an electrical signal by the imaging element, performs noise reduction processing and the like, and outputs image data constituted by a digital signal.

[0039] The control unit **101** performs resizing processing and color conversion processing, such as pixel interpolation and reduction, on image data imaged by the imaging unit **102**. Further, the control unit **101** compression-encodes still image data, which has been subjected to image processing, by using JPEG or the like or encodes moving image data by using a moving image compression method, such as MPEG2 or H.264, and thereby generates an image file and stores it in a recording medium **108**. In the digital camera **100** according to the present embodiment, image data is stored in the recording medium **108** according to a Design rule for Camera File system (DCF) standard. Further, the control unit **101** performs predetermined computational processing by using captured image data and, by controlling the focus lens, a diaphragm, and the shutter of the imaging unit **102** based on the obtained computational result, performs auto focus (AF) processing and auto exposure (AE) processing. The non-volatile memory **103** is an electrically erasable/recordable memory, and for example, an EEPROM or the like is used. In the non-volatile memory **103**, constants, programs, and the like for operation of the control unit **101** are stored. Here, a program is a program for executing control processing, which will be described later in the present embodiment.

[0040] A working memory **104** is used as a work region for loading constants for operation of the control unit **101**, variables, programs read out from the non-volatile memory **103**, and the like. The working memory **104** is used as a buffer memory for temporarily holding image data captured by the imaging unit **102** and an image display memory of a display unit **106**.

[0041] An operation unit **105** is constituted by operation members, such as various switches, buttons, and dials that receive various operations from a user. The operation unit **105** includes, for example, a power button for turning the power on or off, a shutter button for instructing shooting of an image, a reproduction button for instructing reproduction of an image, and a mode switching button for changing the operation mode of the camera. Further, the operation unit **105** includes a dedicated connection button for starting communication with an external apparatus, such as an FTP server **200**, which will be described later, and the like. Furthermore, the operation unit **105** includes a touch panel that is integrally constituted in the display unit **106**, which will be described later.

[0042] The shutter button turns on when operated halfway, a so-called half-press (shooting preparation instruction), and generates a first shutter switch signal SW1. Upon receiving the first shutter switch signal SW1, the control unit **101** controls the imaging unit **102** and thereby starts operation, such as auto focus (AF) processing, auto exposure (AE) processing, auto white balance (AWB) processing, and flash pre-emission (EF) processing. Further, the shutter button turns on when operated fully, a so-called full-press (shooting instruction), and generates a second shutter switch signal SW2. Upon receiving the second shutter switch signal SW2, the control unit **101** starts a series of shooting processing operations from readout of image data from the imaging unit **102** to writing of an image file in the recording medium **108**.

[0043] The display unit **106** performs display of a live view image, display of a captured image, display of characters for interactive operation, and the like. The display unit **106** is a display device, such as a liquid crystal display or an organic EL display, for example. The display unit **106** may be configured to be integrated with the digital camera **100** or may be an external apparatus connected to the digital camera **100**. The digital camera **100** may be connected to the display unit **106** and need only be capable of controlling display of the display unit **106**.

[0044] An audio input unit **107** is one or more microphones incorporated in the digital camera **100** or connected via an audio terminal and converts an analog audio signal generated by collecting audio around the digital camera **100** into a digital signal and outputs it to the control unit **101**. The control unit **101** performs various kinds of audio processing on the digital signal generated by the audio input unit **107** and generates audio data. When shooting a moving image, the control unit **101** generates a moving image file with audio based on audio data generated by the audio input unit **107** and image data generated by the imaging unit **102**. Further, when shooting a still image, the control unit **101** associates audio data generated by the audio input unit **107** with image data generated by the imaging unit **102** and attaches audio file to the image file by using the voice memo function and stores it. The processing for the audio input unit **107** to generate audio data from a sound wave may be executed by other hardware (for example, the control unit **101**) sharing parts of the processing.

[0045] The recording medium **108** stores image files and audio files generated by the control unit **101**. Further, in a reproduction mode, the control unit **101** reads out audio files and image files stored in the recording medium **108**. The recording medium **108** may be a memory card, a hard disk drive, or the like mounted on the digital camera **100** or may be a flash memory or a hard disk drive incorporated in the digital camera **100**. The digital camera **100** need only be capable of accessing at least the recording medium **108**.

[0046] A wireless connection unit **109** includes a communication interface for wirelessly communicating with an external apparatus. The digital camera **100** according to the present embodiment can exchange data with an external apparatus via the wireless connection unit **109** and can transfer, for example, image files and audio files, to an external apparatus via the wireless connection unit **109**. In the present embodiment, the wireless connection unit **109** includes an antenna and a communication circuit for communicating with an external apparatus by wireless LAN according to an IEEE 802.11 standard. The control unit **101** realizes wireless communication with an external apparatus by controlling the wireless connection unit **109**. The communication method is not limited to the wireless LAN and can include a wireless communication interface, such as an infrared communication interface or a wireless USB, for example. The wireless connection unit **109** need not necessarily be incorporated in the digital camera **100**, and the digital camera **100** can be connected to the incorporated or externally attached wireless connection unit **109** and need only include a function of controlling at least the wireless connection unit **109**.

[0047] A short-range communication unit **110** is constituted by, for example, an antenna for wireless communication, a modulation/demodulation circuit for processing wireless signals, and a communication controller. The short-range communication unit **110** realizes short-range wireless communication according to an IEEE 802.15 standard (Bluetooth (registered trademark)) by outputting modulated wireless signals from the antenna and demodulating wireless signals received by the antenna. In the present embodiment, communication in Bluetooth employs Version 4.0 of Bluetooth (registered trademark) Low Energy (BLE), which consumes little power. Communication in Bluetooth (registered trademark) has a narrower range in which communication is possible (i.e., has a shorter range in which communication is possible) than communication in a wireless LAN. Further, communication in Bluetooth (registered trademark) has a slower communication speed than communication in the wireless LAN. Meanwhile, communication in Bluetooth (registered trademark) consumes less power than communication in the wireless LAN. The digital camera **100** according to the present embodiment can exchange data with an external apparatus via the short-range communication unit **110**. Upon receiving a shooting instruction from an external apparatus via the short-range communication unit **110**, for example, the digital camera **100** controls the imaging unit **102** and executes an imaging operation.

[0048] A wired connection unit **111** includes a communication interface for communicating with an external apparatus by wire. The digital camera **100** according to the present embodiment can exchange data with an external apparatus via the wired connection unit **111**. In the present embodiment, the wired connection unit **111** includes a communication circuit for communicating with an external apparatus by wired LAN according to an Ethernet standard. The control unit **101** realizes wired communication with an external apparatus by controlling the wired connection unit **111**. The communication method is not limited to Ethernet and may include USB, HDMI (registered trademark), and IEEE 1394.

[0049] The wireless connection unit **109** of the digital camera **100** according to the present embodiment includes an AP mode in which it operates as an access point in infrastructure mode and a CL mode in which it operates as a client in infrastructure mode. Thus, by operating the wireless connection unit **109** in CL mode, the digital camera **100** according to the present embodiment can operate as a CL device in infrastructure mode. When operating as a CL device, the digital camera **100** can join a network formed by a nearby access point device by connecting to the access point device. Further, by operating the wireless connection unit **109** in AP mode, the digital camera **100** according to the present embodiment can operate as a type of access point, albeit a simple access point (hereinafter, referred to as a simple AP) with limited function. When the digital camera **100** operates as a simple AP, the digital camera **100** forms a network on its own. Apparatuses near the digital camera **100** can recognize the digital camera **100** as an access point device and join the network formed by the digital camera **100**. It is assumed that a program for operating the digital camera **100** as described above is held in the non-volatile memory **103**.

[0050] The digital camera **100** according to the present embodiment is a type of access point, albeit a simple access point that does not have a gateway function for transferring data received from a CL device to an Internet provider or the like. Therefore, even if data is received from another apparatus that has joined the network formed by the digital camera **100**, the data cannot be transferred to a network, such as the Internet.

<Forms of Connection>

[0051] Next, forms of connection between the digital camera **100** and the FTP server **200** in the present embodiment will be described with reference to FIGS. **2**A and **2**B.

[0052] The digital camera **100** according to the present embodiment transmits and receives files to and from the FTP server according to File Transfer Protocol (FTP).

[0053] FIGS. **2**A and **2**B are diagrams illustrating forms of connection between the digital camera **100** and the FTP server **200** according to the present embodiment. FIG. **2**A illustrates a form of connection in which the digital camera **100** and the FTP server **200** connect via an AP **300**.

[0054] In the example of FIG. **2**A, the digital camera **100** and the FTP server **200** join a wireless LAN network formed by the AP **300**, which is an example of an external relay apparatus, the digital camera **100** designates and connects to the FTP server **200**, and the digital camera **100** and the FTP server **200** can thereby communicate by FTP. FIG. **2**B illustrates a form of connection in which the digital camera **100** and the FTP server **200** connect via a hub **400**.

[0055] In the example of FIG. **2**B, the digital camera **100** and the hub **400** of a wired LAN, which is an example of an external relay apparatus, connect by wired LAN cable, the digital camera **100** designates and connects to the FTP server **200**, and the digital camera **100** and the FTP server **200** can thereby communicate by FTP.

[0056] In the present embodiment, an example in which the digital camera **100** is connected with the FTP server **200** will be described, but the present disclosure is not limited thereto. Further, in the present embodiment, an example in which the digital camera **100** and the FTP server **200** are connected to a hub of a wired LAN by wired LAN cable will be described, but the present disclosure is not limited thereto. For example, it is possible to establish a connection with an AP by wired LAN cable or directly establish a connection with the FTP server **200** by wired LAN cable.

<Method of Transferring Files from Digital Camera to FTP Server>

[0057] A method of transferring files from the digital camera **100** to the FTP server **200** according to the present embodiment will be described with reference to FIG. **3**.

[0058] When the digital camera **100** is connected to the FTP server **200**, a plurality of FTP sessions can be established between the digital camera **100** and the FTP server. The digital camera **100** transfers image files in a plurality of FTP sessions with the FTP server **200**. FIG. **3** illustrates a state in which two FTP sessions **301** and **302** have been established between the digital camera **100** and the FTP server **200**. When transferring image files from the digital camera **100** to the FTP server **200**, the next image file is transferred when one FTP session becomes available (when file transfer is completed), and the next image file is transferred when the other FTP session becomes available. The image files are thus transferred in order each time an FTP session becomes available. For example, there is a method of performing data transfer in order in which FTP sessions have become available, such as the first image file in an FTP session **301**, the next image file in an FTP session **302**, and the next image file in the FTP session **301** in which data transfer has been completed. By thus establishing a plurality of FTP sessions, a plurality of image files can be simultaneously transmitted in parallel, and so, it is possible to improve transfer throughput.

[0059] In the following description, it is assumed that an audio file is an audio file generated by the voice memo function.

<Problem that Occurs when Transferring Image File and Audio File Attached to Image File from Digital Camera to FTP Server in Plurality of FTP Sessions>

[0060] Next, a conventional problem that occurs when transferring an image file and an audio file attached to the image file from the digital camera to the FTP server will be described with reference to FIG. **4**.

[0061] FIG. **4** is a diagram illustrating a conventional problem of transferring an image file and an audio file attached to the image file from the digital camera to the FTP server.

[0062] When transferring an image file and an audio file attached to the image file from the digital camera **100** to the

FTP server in a plurality of FTP sessions, the audio file may end up arriving at the FTP server before the image file.

[0063] A user that performs shooting with the digital camera **100** attaches an audio file to an image file and transfers the image file and the audio file to the FTP server. Further, a user that edits image files performs processing such as editing of an image file to which an audio file is attached with arrival of the audio file at the FTP server as a trigger. When such a workflow is assumed, it is necessary to perform control such that the audio file arrives at the FTP server later than the image file.

[0064] In the example of FIG. **4**, two FTP sessions are established between the digital camera **100** and the FTP server, and files, which are XXX1.JPG, XXX1.WAV, XXX2.JPG, XXX3.JPG, XXX3.CR3, XXX3.WAV, XXX4.JPG, and XXX4.WAV, are transferred in order in session 1 and session 2. XXX1.WAV is an audio file attached to XXX1.JPG. XXX3.WAV is an audio file attached to XXX3.JPG and XXX3.CR3. XXX3.CR3 is a RAW file.

[0065] First, XXX1.JPG is transferred in session 1 and XXX1.WAV is transferred in session 2, but since the audio file has a smaller data size than the image file, it is highly likely that XXX1.WAV, which is an audio file, will arrive at the FTP server before XXX1.JPG, which is the image file. Further, regarding image files and an audio file with a filename XXX3, since session 1 becomes available, XXX3.JPG is transferred in session 1, then since session 2 becomes available, XXX3.CR3 is transferred in session 2, and then since session 1 becomes available, XXX3.WAV is transferred in session 1, but it is highly likely that XXX3.WAV, which is the audio file, will arrive at the FTP server before XXX3.CR3, which is a RAW image and is large in size.

[0066] Thus, although it is necessary that the audio file arrives at the FTP server after the image file, when transferring an image file and an audio file in a plurality of FTP sessions, there is a problem that the audio file arrives at the FTP server before the image file.

[0067] Therefore, in the present embodiment, when transferring an image file and an audio file attached to the image file in a plurality of FTP sessions, image files with the same file name are transferred in the same FTP session, and the audio file is transferred last. This makes it possible to perform control such that the audio file arrives at an external apparatus later than the image file.

<Method of Transferring Image File and Audio File Attached to Image File from Digital Camera to FTP Server>

[0068] Next, a method of transferring an image file and an audio file attached to the image file from the digital camera to the FTP server according to Embodiment 1 will be described with reference to FIGS. **5**A to **7**C.

[0069] FIGS. **5**A to **5**F illustrate UI screens for selecting an image file to be transferred from the digital camera to the FTP server according to Embodiment 1. FIG. **6** illustrates a method of transferring an image file and an audio file attached to the image file from the digital camera to the FTP server according to Embodiment 1. FIGS. **7**A to **7**C illustrate control processing for transferring an image file and an audio file attached to the image file from the digital camera to the FTP server according to Embodiment 1.

[0070] The processing of FIGS. **7**A to **7**C is realized by the control unit **101** loading a program stored in the non-volatile memory **103** in the working memory **104** and executing the program, and controlling the respective components of the digital camera **100**. It is similar for FIGS. **9**A and **9**B and FIGS. **11**A and **11**B, which will be described later.

[0071] In step S**701**, the control unit **101** determines whether an image file to be transferred to the FTP server has been selected by a user operating the operation unit **105**. The control unit **101** repeats the processing until it is determined that an image file has been selected and, when it is determined that an image file has been selected, advances the processing to step S**702**.

[0072] Here, UI screens of the digital camera **100** for the user to select an image file to be transferred to the FTP server will be described with reference to FIGS. **5**A to **5**F.

[0073] FIG. **5**A illustrates a network setting screen **501** of the digital camera **100**. The network setting screen **501** allows selection of one selection item among: “network settings”, on/off of “airplane mode”, and “image transfer”. When the user selects an “image transfer” option **501***a* in the screen **501** of FIG. **5**A, an image selection/transfer screen **502** illustrated in FIG. **5**B is transitioned to.

[0074] FIG. **5**B illustrates the screen **502** on which image selection/transfer is performed. In the image selection/transfer screen **502** of FIG. **5**B, a “select image” button, a “designate range” button, a “select folder” button, an “all images” button, and a “transfer” button are displayed. When the user operates the “select image” button in the screen **502** of FIG. **5**B, they can select one image file at a time to be transferred; when they operate the “designate range” button, they can designate a range of image files to be transferred; when they operate the “select folder” button, they can select a folder to be transferred; when they operate the “all images” button, they can select all images; and when they operate the “transfer” button, they can start transfer. When the user operates the “select image” button **502***a* in the screen **502** of FIG. **5**B, a screen **503** illustrated in FIG. **5**C is transitioned to. In the present embodiment, a case where the “select image” button **502***a* is operated in the screen **502** of FIG. **5**B will be described, but the method of selecting an image file is not limited thereto.

[0075] FIG. **5**C illustrates the screen **503** for selecting one image file at a time to be transferred. In the screen **503**, an image file to be transferred can be selected by checking (enabling) a checkbox **503***a* corresponding to the image file to be transferred. When an image file to which an audio file is attached is selected, the audio file is selected as the transfer target with the image file. After image file selection has been completed, when the user operates a “MENU” button **503***b* in the screen **503** of FIG. **5**C, the image selection/transfer screen **502** illustrated in FIG. **5**B is returned to, and when a “SET” button **503***c* is operated, an image selection/transfer screen **504** illustrated in FIG. **5**D is transitioned to. The image selection/transfer screen **504** illustrated in FIG. **5**D is the same as the image selection/transfer screen **502** illustrated in FIG. **5**B but is in a state in which image files to be transferred have been selected.

[0076] FIG. **5**D illustrates the image selection/transfer screen **504** after image files to be transferred have been selected. In the screen **504** of FIG. **5**D, the number of image files to be transferred **504***a* has increased compared with that of the screen **502** of FIG. **5**B before image files to be transferred have been selected. In the screen **504** of FIG. **5**D, when the user operates a “transfer” button **504***b*, a transfer start screen **505** illustrated in FIG. **5**E is transitioned to.

[0077] FIG. **5**E illustrates the screen **505** in which the user can select whether to start transfer image files. By the user operating an "OK" button **505***a* in the transfer start screen **505** of FIG. **5**E, transfer of image files is started, and by them operating a "cancel" button **505***b*, the screen **504** illustrated in FIG. **5**D is returned to without starting transfer of image file.

[0078] In step S**701**, the control unit **101** determines whether an image file has been selected in the screen **503** of FIG. **5**C and the "OK" button **505***a* has been operated in the screen **505** of FIG. **5**E. In the present embodiment, as illustrated in FIG. **6**, it is assumed that a plurality of image files in different file formats are included, the different file formats include at least JPEG or HEIF and RAW, and image files with file numbers XXX1, XXX2, XXX3, XXX4 have been selected. When the user operates the "OK" button **505***a* in the screen **505** of FIG. **5**E, a screen **506** illustrated in FIG. **5**F is transitioned to. The screen **506** illustrated in FIG. **5**F illustrates that the selected image files are being transferred by FTP and displays link rate, remaining transfer time, remaining number of sheets, and the like. When the user operates a "cancel transfer" button **506***a* in the screen **506** of FIG. **5**F, FTP transfer is canceled.

[0079] In step S**702**, the control unit **101** rearranges files with the same file name in order of JPEG/HEIF, RAW, and audio file among the images selected in step S**701**. In the example of FIG. **6**, files with the file name XXX3 are rearranged in order of XXX3.JPG, XXX3.CR3, and XXX3.WAV.

[0080] In step S**703**, the control unit **101** stores in the working memory **104** and registers in a transfer list the image file selected and rearranged in steps S**701**. In the present embodiment, as illustrated in FIG. **6**, it is assumed that XXX1.JPG, XXX1.WAV, XXX2.JPG, XXX3.JPG, XXX3.CR3, XXX3.WAV, XXX4.JPG, and XXX5.JPG are registered in the transfer list in that order.

[0081] In step S**704**, the control unit **101** references the first file in the transfer list registered in step S**703**. In the present embodiment, as illustrated in FIG. **6**, since XXX1.JPG is the first file in the transfer list, processing for transferring XXX1.JPG will be described.

[0082] In step S**705**, the control unit **101** determines whether the first file in the transfer list referenced in step S**704** is an image file. When it is determined to be an image file, the control unit **101** advances the processing to step S**706**, and when it is determined not to be an image file, that is, when it is determined to be an audio file, the control unit **101** advances the processing to step S**721**. Since XXX1.JPG is an image file, the control unit **101** advances the processing to step S**706**.

[0083] In step S**706**, the control unit **101** determines whether an audio file is attached to the image file. When it is determined that an audio file is attached, the control unit **101** advances the processing to step S**707**, and when it is determined that an audio file is not attached, the control unit **101** advances the processing to step S**716**. As illustrated in FIG. **6**, since an audio file XXX1.WAV is attached to XXX1.JPG, the control unit **101** advances the processing to step S**707**.

[0084] In step S**707**, the control unit **101** determines whether the image file is JPEG or HEIF. When it is determined that the image file is JPEG or HEIF, the control unit **101** advances the processing to step S**708**, and when it is determined that the image file is not JPEG or HEIF, that is, RAW, the control unit **101** advances the processing to step S**714**. Since XXX1.JPG is JPEG, the control unit **101** advances the processing to step S**708**.

[0085] In step S**708**, the control unit **101** determines whether there is an available FTP session. The control unit **101** waits until it is determined that there is an available FTP session, and when it is determined that there is an available FTP session, the control unit **101** advances the processing to step S**709**. When XXX1.JPG is referenced in step S**704**, both FTP sessions **1** and **2** are in a state in which no image file is being transferred, and thus, the control unit **101** determines that FTP session 1 illustrated in FIG. **6** is available and advances the processing to step S**709**.

[0086] In step S**709**, the control unit **101** determines whether a file is registered in the transfer pending list of the FTP session determined to be available in step S**708**. When it is determined that a file is registered in the transfer pending list, the control unit **101** advances the processing to step S**710**, and when it is determined that no file is registered in the transfer pending list, the control unit **101** advances the processing to step S**712**. When an FTP session is not available and an image file is being transferred, a file to be transferred next in the same FTP session is registered in the transfer pending list. When the FTP session becomes available, transfer is performed in order from the file registered in the transfer pending list. In the present embodiment, it is assumed that a transfer pending list is stored in the working memory **104** for each FTP session. When XXX1.JPG is referenced, no file is registered in the transfer pending list of FTP session 1, and thus, the control unit **101** advances the processing to step S**712**.

[0087] In step S**712**, the control unit **101** starts transfer of the image file in the FTP session determined to be available in step S**708** and advances the processing to step S**713**. As illustrated in FIG. **6**, the control unit **101** starts transfer of XXX1.JPG in FTP session 1.

[0088] In step S**713**, the control unit **101** stores, in the working memory **104**, the number of the FTP session in which transfer of the image file has been started in step S**712** and advances the processing to step S**722**. The control unit **101** has started transfer of XXX1.JPG in FTP session 1 and thus stores FTP session 1.

[0089] In step S**722**, the control unit **101** deletes the file for which transfer has been started in step S**712** from the transfer list in which the file has been registered in step S**703** and advances the processing to step S**723**. The control unit **101** deletes XXX1.JPG from the transfer list.

[0090] In step S**723**, the control unit **101** determines whether a file remains in the transfer list and, when it is determined that a file remains, returns the processing to step S**704** and, when it is determined that no file remains, advances the processing to step S**724**. The control unit **101** deletes XXX1.JPG from the transfer list but, since files XXX1.WAV to XXX5.JPG remain, returns the processing to step S**704**.

[0091] This concludes the processing for transferring XXX1.JPG.

[0092] Next, in step S**704**, the control unit **101** references the first file in the transfer list again. In this case, the first file in the transfer list is XXX1.WAV.

[0093] In step S**705**, the control unit **101** determines that the first file XXX1.WAV in the transfer list referenced in step S**704** is an audio file and thus advances the processing to step S**721**.

[0094] In step **S721**, the control unit **101** registers the audio file in the transfer pending list of the FTP session stored in step **S713** and advances the processing to step **S722**. Since the FTP session number stored when XXX1.JPG is transferred is FTP session 1, the control unit **101** registers XXX1.WAV in the transfer pending list of FTP session 1.

[0095] In steps **S722** and **S723**, the control unit **101** deletes XXX1.WAV from the transfer list and, since files XXX2.JPG to XXX5.JPG remain in the transfer list, determines that a file remains in the transfer list and returns the processing to step **S704**.

[0096] This concludes the processing for transferring XXX1.WAV.

[0097] In steps **S704** and **S705**, the control unit **101** determines that the first file in the transfer list is XXX2.JPG and is an image file again and thus advances the processing to step **S706**.

[0098] In step **S706**, the control unit **101** determines that an audio file is not attached to the image file XXX2.JPG as illustrated in FIG. **6**, and thus advances the processing to step **S716**.

[0099] In step **S716**, the control unit **101** determines whether there is an available FTP session. The control unit **101** waits until it is determined that there is an available FTP session, and when it is determined that there is an available FTP session, the control unit **101** advances the processing to step **S717**. When XXX2.JPG is referenced in step **S704**, FTP session 2 is in a state in which no image file is being transferred as illustrated in FIG. **6**, and thus, the control unit **101** determines that FTP session 2 is available and advances the processing to step **S717**.

[0100] In step **S717**, the control unit **101** determines whether a file is registered in the transfer pending list of the FTP session determined to be available in step **S716**. When it is determined that a file is registered in the transfer pending list, the control unit **101** advances the processing to step **S718**, and when it is determined that no file is registered in the transfer pending list, the control unit **101** advances the processing to step **S720**. When XXX2.JPG is referenced, no file is registered in the transfer pending list of FTP session 2, and thus, the control unit **101** advances the processing to step **S720**.

[0101] In step **S720**, the control unit **101** starts transfer of the image file in the FTP session determined to be available in step **S716**. As illustrated in FIG. **6**, the control unit **101** transfers XXX2.JPG in FTP session 2.

[0102] In steps **S722** and **S723**, the control unit **101** deletes XXX2.JPG from the transfer list and, since files XXX3.JPG to XXX5.JPG remain in the transfer list, determines that a file remains in the transfer list and returns the processing to step **S704**.

[0103] This concludes the processing for transferring XXX2.JPG.

[0104] In step **S704**, the control unit **101** references the first file in the transfer list again. In this case, the first file in the transfer list is XXX3.JPG. Since XXX3.JPG is an image file and an audio file is attached, the control unit **101** performs processing of steps **S705**, **S706**, **S707**, and **S708**.

[0105] In step **S708**, the control unit **101** determines whether an FTP session is available. When XXX3.JPG is referenced in step **S704**, XXX1.JPG is being transferred in FTP session 1 and XXX2.JPG is being transferred in FTP session 2, and thus, the control unit **101** waits until it is determined that an FTP session is available. Then, the control unit **101** determines that transfer of XXX1.JPG in FTP session 1 has been completed and FTP session 1 has become available and advances the processing to step **S709**.

[0106] In step **S709**, the control unit **101** determines whether a file is registered in a transfer pending list of the available FTP session. When XXX3.JPG is referenced in step **S704**, XXX1.WAV is registered in the transfer pending list of FTP session 1 determined to be available in step **S708**, and thus, the control unit **101** advances the processing to step **S710**.

[0107] In step **S710**, the control unit **101** starts transfer of a file in the transfer pending list in the FTP session determined to be available in step **S708** and advances the processing to step **S711**. In this case, since XXX1.WAV is registered in the transfer pending list of FTP session 1, the control unit **101** starts transfer of XXX1.WAV in FTP session 1.

[0108] In step **S711**, the control unit **101** deletes the file for which transfer has been started in step **S710** from the transfer pending list and returns the processing to step **S708**. The control unit **101** deletes XXX1.WAV for which transfer has been started in step **S710** from the transfer pending list of FTP session 1. In this case, a state in which no file is registered in the transfer pending list of FTP session 1 is entered.

[0109] In step **S708**, the control unit **101** determines whether an FTP session is available again. At this time, XXX1.WAV is being transferred in FTP session 1 and XXX2.JPG is being transferred in FTP session 2, and thus, the control unit **101** waits until it is determined that an FTP session is available. Then, the control unit **101** determines that transfer of XXX2.JPG in FTP session 2 has been completed and FTP session 2 has become available and advances the processing to step **S709**.

[0110] In step **S709**, the control unit **101** determines whether a file is registered in the transfer pending list of the FTP session determined to be available in step **S708**. In this case, since no file is registered in the transfer pending list of FTP session 2 determined to be available in step **S708**, the control unit **101** advances the processing to step **S712**.

[0111] In steps **S712** and **S713**, the control unit **101** starts transfer of XXX3.JPG in FTP session 2 determined to be available in step **S708**, stores FTP session 2 as the FTP session in which transfer has been performed, and advances the processing to step **S722**.

[0112] In steps **S722** and **S723**, the control unit **101** deletes XXX3.JPG from the transfer list and, since files XXX3.CR3 to XXX5.JPG remain in the transfer list, determines that a file remains in the transfer list and returns the processing to step **S704**.

[0113] This concludes the processing for transferring XXX3.JPG.

[0114] In step **S704**, the control unit **101** references the first file in the transfer list again. In this case, the first file in the transfer list is XXX3.CR3. Since XXX3.CR3 is an image file and an audio file is attached, the control unit **101** performs processing of steps **S705**, **S706**, and **S707**.

[0115] In step **S707**, the control unit **101** determines that XXX3.CR3 is a RAW file and advances the processing to step **S714**.

[0116] In step **S714**, the control unit **101** determines whether there is JPEG/HEIF with the same file names as the image file. When it is determined that there is JPEG/HEIF with the same file name, the control unit **101** advances the processing to step **S715**, and when it is determined that there is no JPEG/HEIF with the same file name, the control unit **101** advances the processing to step **S715**. Since XXX3.CR3 has XXX3.JPG with the same file name, the control unit **101** advances the processing to step **S715**.

[0117] In step **S715**, the control unit **101** registers the image file in the transfer pending list of the FTP session stored in step **S713** and advances the processing to step **S722**. Since the FTP session number stored when XXX3.JPG is transferred is FTP session 2, the control unit **101** registers XXX3.CR3 in the transfer pending list of FTP session 2.

[0118] In steps **S722** and **S723**, the control unit **101** deletes XXX3.CR3 from the transfer list and, since files XXX3.WAV to XXX5.JPG remain in the transfer list, determines that a file remains in the transfer list and returns the processing to step **S704**.

[0119] This concludes the processing for transferring XXX3.CR3.

[0120] In steps **S704** and **S705**, the control unit **101** determines that the first file in the transfer list is XXX3.WAV and is an audio file and thus advances the processing to step **S721**.

[0121] In step **S721**, the control unit **101** registers the audio file in the transfer pending list of the FTP session stored in step **S713** and advances the processing to step **S722**. Since the FTP session number stored when XXX3.JPG is transferred is FTP session 2, the control unit **101** registers XXX3.WAV in the transfer pending list of FTP session 2. In this case, the state is such that XXX3.CR3 and XXX3.WAV are registered in the transfer pending list of FTP session 2.

[0122] In steps **S722** and **S723**, the control unit **101** deletes XXX3.WAV from the transfer list and, since files XXX4.JPG and XXX5.JPG remain in the transfer list, determines that a file remains in the transfer list and returns the processing to step **S704**.

[0123] This concludes the processing for transferring XXX3.WAV.

[0124] In step **S704**, the control unit **101** references the first file in the transfer list again. In this case, the first file in the transfer list is XXX4.JPG. Since XXX4.JPG is an image file and no audio file is attached, the control unit **101** performs processing of steps **S705** and **S716**.

[0125] In step **S716**, the control unit **101** determines whether an FTP session is available. When XXX4.JPG is referenced in step **S704**, XXX1.WAV is being transferred in FTP session 1 and XXX3.JPG is being transferred in FTP session 2, and thus, the control unit **101** waits until it is determined that an FTP session is available. Then, the control unit **101** determines that transfer of XXX1.WAV in FTP session 1 has been completed and FTP session 1 has become available and advances the processing to step **S717**.

[0126] In step **S717**, the control unit **101** determines whether a file is registered in the transfer pending list of the FTP session determined to be available in step **S716**. When XXX4.JPG is referenced in step **S704**, no file is registered in the transfer pending list of FTP session 1 determined to be available in step **S716**, and thus, the control unit **101** advances the processing to step **S720**.

[0127] In step **S720**, the control unit **101** starts transfer of the image file in the FTP session determined to be available in step **S716** and advances the processing to step **S722**. As illustrated in FIG. **6**, the control unit **101** starts transfer of XXX4.JPG in FTP session 1.

[0128] In steps **S722** and **S723**, the control unit **101** deletes XXX4.JPG from the transfer list and, since XXX5.JPG remains in the transfer list, determines that a file remains in the transfer list and returns the processing to step **S704**.

[0129] This concludes the processing for transferring XXX4.JPG.

[0130] In step **S704**, the control unit **101** references the first file in the transfer list again. In this case, the first file in the transfer list is XXX5.JPG. Since XXX5.JPG is an image file and no audio file is attached, the control unit **101** performs processing of steps **S705** and **S716**.

[0131] In step **S716**, the control unit **101** determines whether there is an available FTP session. When XXX5.JPG is referenced in step **S704**, XXX4.JPG is being transferred in FTP session 1 and XXX3.JPG is being transferred in FTP session 2, and thus, the control unit **101** waits until it is determined that there is an available FTP session. Then, the control unit **101** determines that transfer of XXX3.JPG in FTP session 2 has been completed and FTP session 2 has become available and advances the processing to step **S717**.

[0132] In step **S717**, the control unit **101** determines whether a file is registered in the transfer pending list of the FTP session determined to be available in step **S716**. When XXX5.JPG is referenced in step **S704**, XXX3.CR3 and XXX3.WAV are registered in the transfer pending list of FTP session 2 determined to be available in step **S716**, and thus, the processing proceeds step **S718**.

[0133] In step **S718**, the control unit **101** starts transfer of a file in the transfer pending list in the FTP session determined to be available in step **S716** and advances the processing to step **S719**. In this case, since XXX3.CR3 and XXX3.WAV are registered in the transfer pending list of FTP session 2, the control unit **101** starts transfer of XXX3.CR3 in FTP session 2.

[0134] In step **S719**, the control unit **101** deletes the file for which transfer has been started in step **S718** from the transfer pending list and returns the processing to step **S716**. The control unit **101** deletes XXX3.CR3 for which transfer has been started in step **S718** from the transfer pending list of FTP session 2. In this case, the state is such that XXX3.WAV remains in the transfer pending list.

[0135] In step **S716**, the control unit **101** determines whether an FTP session is available. At this time, XXX4.JPG is being transferred in FTP session 1 and XXX3.CR3 is being transferred in FTP session 2, and thus, the control unit **101** waits until it is determined that an FTP session is available. Then, the control unit **101** determines that transfer of XXX4.JPG in FTP session 1 has been completed and FTP session 1 has become available and advances the processing to step **S717**.

[0136] In step **S717**, the control unit **101** determines whether a file is registered in the transfer pending list of the FTP session determined to be available in step **S716**. At this time, no file is registered in the transfer pending list of FTP session 1 determined to be available in step **S716**, and thus, the processing proceeds to step **S720**.

[0137] In step **S720**, the control unit **101** starts transfer of the file in the FTP session determined to be available in step **S716** and advances the processing to step **S722**. As illustrated in FIG. **6**, the control unit **101** starts transfer of XXX5.JPG in FTP session 1.

[0138] In steps S**722** and S**723**, the control unit **101** deletes XXX5.JPG from the transfer list and, since no file remains in the transfer list, determines that no file remains in the transfer list and advances the processing to step S**724**.

[0139] This concludes the processing for transferring XXX5.JPG.

[0140] In step S**724**, the control unit **101** determines whether a file remains in the transfer pending list of FTP session 1 or 2. When it is determined that a file remains in a transfer pending list, the control unit **101** advances the processing to step S**725**, and when it is determined that no file remains in a transfer pending list, the control unit **101** ends the processing. In this case, since XXX3.WAV remains in the transfer pending list of FTP session 2, the control unit **101** advances the processing to step S**725**.

[0141] In step S**725**, the control unit **101** determines whether there is an available FTP session for which it has been determined that a file remains in the transfer pending list in step S**724**. The control unit **101** waits until it is determined that there is an available FTP session, and when it is determined that there is an available FTP session, the control unit **101** advances the processing to step S**726**. In this case, since FTP session 2 is the one in which a file remains in the transfer pending list, and XXX3.CR3 is being transferred in FTP session 2, the control unit **101** waits until it is determined that FTP session 2 is available. Then, the control unit **101** determines that transfer of XXX3.CR3 in FTP session 2 has been completed and FTP session 2 has become available and advances the processing to step S**726**.

[0142] In step S**726**, the control unit **101** starts transfer of a file registered in the transfer pending list and advances the processing to step S**727**. The control unit **101** starts transfer of XXX3.WAV registered in the transfer pending list of FTP session 2 in FTP session 2.

[0143] In step S**727**, the control unit **101** deletes the file for which transfer has been started in step S**726** from the transfer pending list and returns the processing to step S**724**. The control unit **101** deletes XXX3.WAV registered in the transfer pending list of FTP session 2, and the state in which no file is registered in the transfer pending list is entered.

[0144] In step S**724**, the control unit **101** determines whether a file remains in the transfer pending list of FTP session 1 or 2 and, since no file remains in the transfer pending lists, ends the processing.

[0145] As described above, according to Embodiment 1, when transferring an image file and an audio file in a plurality of FTP sessions, by transferring image files with the same file name in the same FTP session and transferring the audio file last, it is possible to perform control such that the audio file arrives at the FTP server after the image file.

Embodiment 2

[0146] Next, Embodiment 2 will be described with reference to FIG. **8** and FIGS. **9**A and **9**B.

[0147] In Embodiment 2, processing in which when the digital camera **100** establishes a plurality of FTP sessions with the FTP server **200** and transfers an image file and an audio file attached to the image file, when there are two files with the same file name including the audio file, the audio file is transferred last in the same FTP session, and when there are three files with the same file name including the audio file, JPEG/HEIF and RAW are transferred in available FTP sessions in that order and the audio file is transferred in the same FTP session as RAW will be described.

[0148] FIG. **8** illustrates a method of transferring image files and audio files attached to the image files from the digital camera to the FTP server according to Embodiment 2. FIGS. **9**A and **9**B illustrate control processing for transferring an image file and an audio file attached to the image file from the digital camera to the FTP server according to Embodiment 2.

[0149] Steps S**901** to S**903** of FIG. **9**A are similar to steps S**701** to S**703** of FIG. **7**A.

[0150] In the present embodiment, as illustrated in FIG. **8**, it is assumed that XXX1.JPG, XXX1.WAV, XXX2.JPG, XXX3.JPG, XXX3.CR3, XXX3.WAV, XXX4.JPG, and XXX5.JPG are registered in the transfer list in that order.

[0151] In step S**904**, the control unit **101** references the first file in the transfer list registered in step S**903** and advances the processing to step S**905**. As illustrated in FIG. **8**, since XXX1.JPG is the first file in the transfer list, processing for XXX1.JPG will be described.

[0152] In step S**905**, the control unit **101** determines whether the first file in the transfer list referenced in step S**904** is an image file. When it is determined to be an image file, the control unit **101** advances the processing to step S**906**, and when it is determined not to be an image file, that is, when it is determined to be an audio file, the control unit **101** advances the processing to step S**915**. Since XXX1.JPG is an image file, the control unit **101** advances the processing to step S**906**.

[0153] In step S**906**, the control unit **101** determines whether there is an available FTP session. The control unit **101** waits until it is determined that there is an available FTP session, and when it is determined that there is an available FTP session, the control unit **101** advances the processing to step S**907**. When XXX1.JPG is referenced in step S**904**, both FTP sessions **1** and **2** are in a state in which no image file is being transferred, and thus, the control unit **101** determines that FTP session 1 illustrated in FIG. **8** is available and advances the processing to step S**907**.

[0154] In step S**907**, the control unit **101** determines whether a file is registered in the transfer pending list of the FTP session determined to be available in step S**906**. When it is determined that a file is registered in the transfer pending list, the control unit **101** advances the processing to step S**908**, and when it is determined that no file is registered in the transfer pending list, the control unit **101** advances the processing to step S**910**. The transfer pending list is similar to that of Embodiment 1. When XXX1.JPG is referenced in step S**904**, no file is registered in the transfer pending list of FTP session 1, and thus, the control unit **101** advances the processing to step S**910**.

[0155] In step S**910**, the control unit **101** starts transfer of the file in the FTP session determined to be available in step S**906** and advances the processing to step S**911**. As illustrated in FIG. **8**, the control unit **101** starts transfer of XXX1.JPG in FTP session 1.

[0156] In step S**911**, the control unit **101** determines whether an audio file is attached to the image file for which transfer has been started in step S**910**. When it is determined that an audio file is attached to the image file, the control unit **101** advances the processing to step S**912**, and when it is determined that no audio file is attached to the image file, the control unit **101** advances the processing to step S**916**. As illustrated in FIG. **8**, since an audio file XXX1.WAV is attached to XXX1.JPG, the control unit **101** advances the processing to step S**912**.

[0157] In step **S912**, the control unit **101** determines whether the image file for which transfer has been started in step **S910** is JPEG or HEIF. When it is determined that the image file is JPEG or HEIF, the control unit **101** advances the processing to step **S913**, and when it is determined that the image file is not JPEG or HEIF, that is, RAW, the control unit **101** advances the processing to step **S914**. Since XXX1.JPG is JPEG, the control unit **101** advances the processing to step **S913**.

[0158] In step **S913**, the control unit **101** determines whether there is RAW with the same file name as the image file for which transfer has been started in step **S910**. When it is determined that there is RAW with the same file name, the control unit **101** advances the processing to step **S916**, and when it is determined that there is no RAW with the same file name, the control unit **101** advances the processing to step **S914**. In this case, since XXX1.JPG has no RAW with the same file name, the control unit **101** advances the processing to step **S914**.

[0159] In step **S914**, the control unit **101** stores, in the working memory **104**, the number of the FTP session in which transfer of the image file has been started in step **S910** and advances the processing to step **S916**. The control unit **101** has started transfer of XXX1.JPG in FTP session 1 and thus stores FTP session 1.

[0160] In step **S916**, the control unit **101** deletes the image file from the transfer list and advances the processing to step **S917**. The control unit **101** deletes XXX1.JPG from the transfer list.

[0161] In step **S917**, the control unit **101** determines whether a file remains in the transfer list and, when it is determined that a file remains, returns the processing to step **S904** and, when it is determined that no file remains, advances the processing to step **S918**. In this case, the control unit **101** deletes XXX1.JPG from the transfer list but, since files XXX1.WAV to XXX5.JPG remain, returns the processing to step **S904**.

[0162] This concludes the processing for transferring XXX1.JPG.

[0163] In steps **S904** and **S905**, the control unit **101** determines that the first file in the transfer list is XXX1.WAV and is an audio file again and thus advances the processing to step **S915**.

[0164] In step **S915**, the control unit **101** registers the audio file in the transfer pending list of the FTP session stored in step **S914** and advances the processing to step **S916**. Since the FTP session number stored when XXX1.JPG is transferred is FTP session 1, the control unit **101** registers XXX1.WAV in the transfer pending list of FTP session 1.

[0165] In steps **S916** and **S917**, the control unit **101** deletes XXX1.WAV from the transfer list and, since files XXX2.JPG to XXX5.JPG remain in the transfer list, determines that a file remains in the transfer list and returns the processing to step **S904**.

[0166] This concludes the processing for transferring XXX1.WAV.

[0167] In steps **S904** and **S905**, the control unit **101** determines that the first file in the transfer list is XXX2.JPG and is an image file again and thus advances the processing to step **S906**.

[0168] In step **S906**, the control unit **101** determines whether an FTP session is available. When XXX2.JPG is referenced in step **S904**, FTP session 2 is in a state in which no image file is being transferred as illustrated in FIG. **8**, and thus, the control unit **101** determines that FTP session 2 is available and advances the processing to step **S907**.

[0169] In step **S907**, the control unit **101** determines whether a file is registered in the transfer pending list of the FTP session determined to be available in step **S906**. When XXX2.JPG is referenced in step **S904**, no file is registered in the transfer pending list of FTP session 2, and thus, the control unit **101** advances the processing to step **S910**.

[0170] In step **S910**, the control unit **101** starts transfer of the image file in the FTP session determined to be available in step **S906** and advances the processing to step **S911**. As illustrated in FIG. **8**, the control unit **101** transfers XXX2.JPG in FTP session 2.

[0171] In step **S911**, the control unit **101** determines that an audio file is not attached to the image file XXX2.JPG for which transfer has been started in step **S910** and thus advances the processing to step **S916**.

[0172] In steps **S916** and **S917**, the control unit **101** deletes XXX2.JPG from the transfer list and, since files XXX3.JPG to XXX5.JPG remain in the transfer list, determines that a file remains in the transfer list and returns the processing to step **S904**.

[0173] This concludes the processing for transferring XXX2.JPG.

[0174] In steps **S904** and **S905**, the control unit **101** determines that the first file in the transfer list is XXX3.JPG and is an image file again and thus advances the processing to step **S906**.

[0175] In step **S906**, the control unit **101** determines whether an FTP session is available. When XXX3.JPG is referenced in step **S904**, XXX1.JPG is being transferred in FTP session 1 and XXX2.JPG is being transferred in FTP session 2, and thus, the control unit **101** waits until it is determined that an FTP session is available. Then, the control unit **101** determines that transfer of XXX1.JPG in FTP session 1 has been completed and FTP session 1 has become available and advances the processing to step **S907**.

[0176] In step **S907**, the control unit **101** determines whether a file is registered in the transfer pending list of the FTP session determined to be available in step **S906**. Since XXX1.WAV is registered in the transfer pending list of FTP session 1, the control unit **101** advances the processing to step **S908**.

[0177] In step **S908**, the control unit **101** starts transfer of a file in the transfer pending list in the FTP session determined to be available in step **S906** and advances the processing to step **S909**. Since XXX1.WAV is registered in the transfer pending list of FTP session 1, the control unit **101** starts transfer of XXX1.WAV in FTP session 1.

[0178] In step **S909**, the control unit **101** deletes the file for which transfer has been started in step **S908** from the transfer pending list and returns the processing to step **S906**. The control unit **101** deletes XXX1.WAV for which transfer has been started in step **S908** from the transfer pending list of FTP session 1. In this case, a state in which no file is registered in the transfer pending list of FTP session 1 is entered.

[0179] In step **S906**, the control unit **101** determines whether an FTP session is available. At this time, XXX1.WAV is being transferred in FTP session 1 and XXX2.JPG is being transferred in FTP session 2, and thus, the control unit **101** waits until it is determined that an FTP session is available. Then, the control unit **101** determines that transfer of XXX2.JPG in FTP session 2 has been completed and FTP session 2 has become available and advances the processing to step S**907**.

[0180] In step S**907**, the control unit **101** determines whether a file is registered in the transfer pending list of the FTP session determined to be available in step S**906**. Here, since no file is registered in the transfer pending list of FTP session 2, the processing proceeds to step S**910**.

[0181] In step S**910**, the control unit **101** starts transfer of the image file in the FTP session determined to be available in step S**906** and advances the processing to step S**911**. As illustrated in FIG. **8**, the control unit **101** transfers XXX3.JPG in FTP session 2.

[0182] In step S**911**, the control unit **101** determines whether an audio file is attached to the image file for which transfer has been started in step S**910**. As illustrated in FIG. **8**, since an audio file XXX3.WAV is attached to XXX3.JPG, the control unit **101** advances the processing to step S**912**.

[0183] In step S**912**, the control unit **101** determines whether the image file for which transfer has been started in step S**910** is JPEG or HEIF. Since XXX3.JPG is JPEG, the control unit **101** advances the processing to step S**913**.

[0184] In step S**913**, the control unit **101** determines whether there is RAW with the same file name as the image file for which transfer has been started in step S**910**. Since XXX3.JPG has RAW with the same file name, which is XXX3.CR3, the control unit **101** advances the processing to step S**916**.

[0185] In steps S**916** and S**917**, the control unit **101** deletes XXX3.JPG from the transfer list and, since files XXX3.CR3 to XXX5.JPG remain in the transfer list, determines that a file remains in the transfer list and returns the processing to step S**904**.

[0186] This concludes the processing for transferring XXX3.JPG.

[0187] In steps S**904** and S**905**, the control unit **101** determines that the first file in the transfer list is XXX3.CR3 and is an image file again and thus advances the processing to step S**906**.

[0188] In step S**906**, the control unit **101** determines whether an FTP session is available. When XXX3.CR3 is referenced in step S**904**, XXX1.WAV is being transferred in FTP session 1 and XXX3.JPG is being transferred in FTP session 2, and thus, the control unit **101** waits until it is determined that an FTP session is available. Then, the control unit **101** determines that transfer of XXX1.WAV in FTP session 1 has been completed and FTP session 1 has become available and advances the processing to step S**907**.

[0189] In step S**907**, the control unit **101** determines whether a file is registered in the transfer pending list of the FTP session determined to be available in step S**906**. Since no file is registered in the transfer pending list of FTP session 1, the control unit **101** advances the processing to step S**910**.

[0190] In step S**910**, the control unit **101** starts transfer of the image file in the session determined to be available in step S**906** and advances the processing to step S**911**. As illustrated in FIG. **8**, the control unit **101** transfers XXX3.CR3 in FTP session 1.

[0191] In step S**911**, the control unit **101** determines whether an audio file is attached to the image file for which transfer has been started in step S**910**. As illustrated in FIG. **8**, since an audio file XXX3.WAV is attached to XXX3.CR3, the control unit **101** advances the processing to step S**912**.

[0192] In step S**912**, the control unit **101** determines whether the image file for which transfer has been started in step S**910** is JPEG or HEIF. Since XXX3.CR3 is RAW, the control unit **101** advances the processing to step S**914**.

[0193] In step S**914**, the control unit **101** stores, in the working memory **104**, the number of the FTP session in which transfer of the image file has been started in step S**910** and advances the processing to step S**916**. The control unit **101** has started transfer of XXX3.CR3 in FTP session 1 and thus stores FTP session 1.

[0194] In steps S**916** and S**917**, the control unit **101** deletes XXX3.CR3 from the transfer list and, since files XXX3.WAV to XXX5.JPG remain in the transfer list, determines that a file remains in the transfer list and returns the processing to step S**904**.

[0195] This concludes the processing for transferring XXX3.CR3.

[0196] In steps S**904** and S**905**, the control unit **101** determines that the first file in the transfer list is XXX3.WAV and is an audio file again and thus advances the processing to step S**915**.

[0197] In step S**915**, the control unit **101** registers the audio file in the transfer pending list of the FTP session stored in step S**914** and advances the processing to step S**916**. Since the number of the FTP session in which transfer of XXX3.CR3 has been started is FTP session 1, the control unit **101** registers XXX3.WAV in the transfer pending list of FTP session 1.

[0198] In steps S**916** and S**917**, the control unit **101** deletes XXX3.WAV from the transfer list and, since files XXX4.JPG and XXX5.JPG remain in the transfer list, determines that a file remains in the transfer list and returns the processing to step S**904**.

[0199] This concludes the processing for transferring XXX3.WAV.

[0200] In steps S**904** and S**905**, the control unit **101** determines that the first file in the transfer list is XXX4.JPG and is an image file again and thus advances the processing to step S**906**.

[0201] In step S**906**, the control unit **101** determines whether an FTP session is available. When XXX4.JPG is referenced in step S**904**, XXX3.CR3 is being transferred in FTP session 1 and XXX3.JPG is being transferred in FTP session 2, and thus, the control unit **101** waits until it is determined that an FTP session is available. Then, the control unit **101** determines that transfer of XXX3.JPG in FTP session 2 has been completed and FTP session 2 has become available and advances the processing to step S**907**.

[0202] In step S**907**, the control unit **101** determines whether a file is registered in the transfer pending list of the FTP session determined to be available in step S**906**. Since no file is registered in the transfer pending list of FTP session 2, the control unit **101** advances the processing to step S**910**.

[0203] In step S**910**, the control unit **101** starts transfer of the image file in the session determined to be available in step S**906** and advances the processing to step S**911**. As illustrated in FIG. **8**, the control unit **101** transfers XXX4.JPG in FTP session 2.

[0204] In step S**911**, the control unit **101** determines whether an audio file is attached to the image file for which transfer has been started in step S**910**. As illustrated in FIG. **8**, since no audio file is attached to XXX4.JPG, the control unit **101** advances the processing to step S**916**.

[0205] In steps S**916** and S**917**, the control unit **101** deletes XXX4.JPG from the transfer list and, since XXX5.JPG remains in the transfer list, determines that a file remains in the transfer list and returns the processing to step S**904**.

[0206] This concludes the processing for transferring XXX4.JPG.

[0207] In steps S**904** and S**905**, the control unit **101** determines that the first file in the transfer list is XXX5.JPG and is an image file again and thus advances the processing to step S**906**.

[0208] In step S**906**, the control unit **101** determines whether an FTP session is available. When XXX5.JPG is referenced in step S**904**, XXX3.CR3 is being transferred in FTP session 1 and XXX4.JPG is being transferred in FTP session 2, and thus, the control unit **101** waits until it is determined that an FTP session is available. Then, the control unit **101** determines that transfer of XXX4.JPG in FTP session 2 has been completed and FTP session 2 has become available and advances the processing to step S**907**.

[0209] In step S**907**, the control unit **101** determines whether a file is registered in the transfer pending list of the FTP session determined to be available in step S**906**. Since no file is registered in the transfer pending list of FTP session 2, the control unit **101** advances the processing to step S**910**.

[0210] In step S**910**, the control unit **101** starts transfer of the image file in the session determined to be available in step S**906** and advances the processing to step S**911**. As illustrated in FIG. **8**, the control unit **101** transfers XXX5.JPG in FTP session 2.

[0211] In step S**911**, the control unit **101** determines whether an audio file is attached to the image file for which transfer has been started in step S**910**. As illustrated in FIG. **8**, since no audio file is attached to XXX5.JPG, the control unit **101** advances the processing to step S**916**.

[0212] In steps S**916** and S**917**, the control unit **101** deletes XXX5.JPG from the transfer list, determines that no file remains in the transfer list, and advances the processing to step S**918**.

[0213] This concludes the processing for transferring XXX5.JPG.

[0214] In step S**918**, the control unit **101** determines whether a file remains in the transfer pending list of FTP session 1 or 2. When it is determined that a file remains in the transfer pending list of FTP session 1 or 2, the control unit **101** advances the processing to step S**919**, and when it is determined that no file remains in the transfer pending lists, the control unit **101** ends the processing. At this time, XXX3.WAV remains in the transfer pending list of FTP session 1, and thus, the control unit **101** advances the processing to step S**919**.

[0215] Steps S**919** to S**921** are similar to steps S**725** to S**727** of FIG. **7**C. Since XXX3.WAV remains in the transfer pending list of FTP session 1, XXX3.WAV is transferred in FTP session 1 when FTP session 1 becomes available. Then, XXX3.WAV is deleted from the transfer pending list of FTP session 1, and the processing is returned to step S**918**.

[0216] In step S**918**, the control unit **101** determines whether a file remains in the transfer pending list of FTP session 1 or 2 and, since no file remains in the transfer pending lists, ends the processing.

[0217] As described above, according to Embodiment 2, when transferring an image file and an audio file attached to the image file in a plurality of FTP sessions, when there are two files with the same file name including the audio file, the audio file is transferred last in the same FTP session, and when there are three files with the same file name including the audio file, JPEG/HEIF and RAW are transferred in available FTP sessions in that order and the audio file is transferred in the same FTP session as RAW. This makes it possible to perform control such that when transferring an image file and an audio file in a plurality of FTP sessions, the audio file arrives at the FTP server later than the image file.

[0218] Since RAW is greater in image size than JPEG/HEIF, when transfer is started with JPEG/HEIF and then RAW, by the time transfer of RAW is completed, transfer of JPEG/HEIF is completed, and the audio file is transferred in the same FTP session as RAW. This makes it possible to perform control such that the audio file arrives at the FTP server later than the image file.

[0219] In the present embodiment, JPEG/HEIF and RAW are transferred in available FTP sessions in that order and the audio file is transferred in the same FTP session as RAW, but a configuration may be taken so as to transfer a plurality of image files in available FTP sessions in order from the file with the smallest size and transfer the audio file in the same FTP session as the image file with the largest size. Thus, for example, even where files with the same file name are a combination of a plurality of types of files, such as JPEG, HEIF, and audio file, by performing transfer in an available FTP session in order from the image file with the smaller size and transferring the audio file in the same FTP session as the image file with the largest size, it is possible to perform control such that when transferring an image file and an audio file attached to the image file in a plurality of FTP sessions, the audio file arrives at the FTP server later than the image file.

Embodiment 3

[0220] Next, Embodiment 3 will be described with reference to FIG. **10** and FIGS. **11**A and **11**B.

[0221] In Embodiment 3, processing in which when the digital camera **100** establishes a plurality of FTP sessions with the FTP server **200** and transfers an image file and an audio file attached to the image file, when there are two files with the same file name including the audio file, the audio file is transferred last in the same FTP session, and when there are three files with the same file name including the audio file, RAW and JPEG/HEIF are transferred in available FTP sessions in that order and the audio file is transferred in the same FTP session as the file for which transfer has been completed last among RAW and JPEG/HEIF will be described.

[0222] FIG. **10** illustrates a method of transferring an image file and an audio file attached to the image file from the digital camera to the FTP server according to Embodiment 3. FIGS. **11**A and **11**B illustrate control processing for transferring an image file and an audio file attached to the image file from the digital camera to the FTP server according to Embodiment 3.

[0223] Steps S**1101** to S**1103** of FIG. **11**A are similar to steps S**701** to S**703** of FIG. **7**A.

[0224] In the present embodiment, as illustrated in FIG. **10**, it is assumed that XXX1.JPG, XXX1.WAV, XXX2.JPG, XXX3.CR3, XXX3.JPG, XXX3.WAV, XXX4.JPG, and XXX4.WAV are registered in the transfer list in that order.

[0225] In step S**1104**, the control unit **101** references the first file in the transfer list registered in step S**1103** and advances the processing to step S**1105**. As illustrated in FIG.

10, since XXX1.JPG is the first file in the transfer list, processing for transferring XXX1.JPG will be described.

[0226] In step S1105, the control unit 101 determines whether the first file in the transfer list referenced in step S1104 is an image file. When it is determined to be an image file, the control unit 101 advances the processing to step S1106, and when it is determined not to be an image file, that is, when it is determined to be an audio file, the control unit 101 advances the processing to step S1112. Since XXX1.JPG is an image file, the control unit 101 advances the processing to step S1106.

[0227] In step S1106, the control unit 101 determines whether there is an available FTP session. The control unit 101 waits until it is determined that there is an available FTP session, and when it is determined that there is an available FTP session, the control unit 101 advances the processing to step S1107. When XXX1.JPG is referenced in step S1104, both FTP sessions 1 and 2 are in a state in which no image file is being transferred, and thus, the control unit 101 determines that FTP session 1 illustrated in FIG. 10 is available and advances the processing to step S1107.

[0228] In step S1107, the control unit 101 determines whether a file is registered in a transfer completion wait list. When it is determined that a file is registered in the transfer completion wait list, the control unit 101 advances the processing to step S1108, and when it is determined that no file is registered in the transfer completion wait list, the control unit 101 advances the processing to step S1111. An audio file is registered in the transfer completion wait list, and when all files with the same file name as the audio file have been transferred, the audio file in the transfer completion wait list is transferred. When XXX1.JPG is selected, since no audio file is registered in the transfer completion wait list, the control unit 101 advances the processing to step S1111.

[0229] In step S1111, the control unit 101 starts transfer of the image file in the FTP session determined to be available in step S1106 and advances the processing to step S1113. As illustrated in FIG. 10, the control unit 101 transfers XXX1.JPG in FTP session 1.

[0230] In step S1113, the control unit 101 deletes the image file from the transfer list and advances the processing to step S1114. The control unit 101 deletes XXX1.JPG from the transfer list.

[0231] In step S1114, the control unit 101 determines whether a file remains in the transfer list. When it is determined that a file remains in the transfer list, the control unit 101 returns the processing to step S1104, and when it is determined that no file remains in a the transfer list, the control unit 101 advances the processing to step S1115. The control unit 101 deletes XXX1.JPG from the transfer list but, since files XXX1.WAV to XXX4.WAV remain, returns the processing to step S1104.

[0232] This concludes the processing for transferring XXX1.JPG.

[0233] In steps S1104 and S1105, the control unit 101 determines that the first file in the transfer list is XXX1.WAV and is an audio file and thus advances the processing to step S1112.

[0234] In step S1112, the control unit 101 registers the audio file in the transfer completion wait list and advances the processing to step S1113. The control unit 101 registers XXX1.WAV in the transfer completion wait list.

[0235] In steps S1113 and S1114, the control unit 101 deletes XXX1.WAV from the transfer list and, since files XXX2.JPG to XXX4.WAV remain in the transfer list, determines that a file remains in the transfer list and returns the processing to step S1104.

[0236] This concludes the processing for transferring XXX1.WAV.

[0237] In steps S1104 and S1105, the control unit 101 determines that the first file in the transfer list is XXX2.JPG and is an image file and thus advances the processing to step S1106.

[0238] In step S1106, the control unit 101 determines whether an FTP session is available. When XXX2.JPG is referenced, FTP session 2 illustrated in FIG. 10 is in a state in which no image file is being transferred, and thus, the control unit 101 determines that FTP session 2 is available and advances the processing to step S1107.

[0239] In step S1107, the control unit 101 determines whether an audio file is registered in the transfer completion wait list. When XXX2.JPG is referenced in step S1104, XXX1.WAV is registered in the transfer completion wait list, and thus, the control unit 101 advances the processing to step S1108.

[0240] In step S1108, the control unit 101 determines whether transfer of all image files with the same file number as the audio file in the transfer completion wait list has been completed. When it is determined that transfer of all image files with the same file number has been completed, the control unit 101 advances the processing to step S1109, and when it is determined that transfer of all image files with the same file number has not been completed, the control unit 101 advances the processing to step S1111. In this case, since XXX1.WAV is registered in the transfer completion wait list and transfer of XXX1.JPG with the same filename has not been completed, the control unit 101 advances the processing to step S1111.

[0241] In step S1111, the control unit 101 starts transfer of the image file in the FTP session determined to be available in step S1106 and advances the processing to step S1113. As illustrated in FIG. 10, the control unit 101 starts transfer of XXX2.JPG in FTP session 2.

[0242] In steps S1113 and S1114, the control unit 101 deletes XXX2.JPG from the transfer list and, since files XXX3.JPG to XXX4.WAV remain in the transfer list, determines that a file remains in the transfer list and returns the processing to step S1104.

[0243] This concludes the processing for transferring XXX2.JPG.

[0244] In steps S1104 and S1105, the control unit 101 determines that the first file in the transfer list is XXX3.CR3 and is an image file and thus advances the processing to step S1106.

[0245] In step S1106, the control unit 101 determines whether an FTP session is available. When XXX3.CR3 is referenced in step S1104, XXX1.JPG is being transferred in FTP session 1 and XXX2.JPG is being transferred in FTP session 2, and thus, the control unit 101 waits until it is determined that an FTP session is available. Then, the control unit 101 determines that transfer of XXX1.JPG in FTP session 1 has been completed and FTP session 1 has become available and advances the processing to step S1107.

[0246] In step S1107, the control unit 101 determines whether an audio file is registered in the transfer completion wait list. When XXX3.CR3 is referenced in step **S1104**, XXX1.WAV is registered in the transfer completion wait list, and thus, the control unit **101** advances the processing to step **S1108**.

[0247] In step **S1108**, the control unit **101** determines whether transfer of all image files with the same file number as the audio file in the transfer completion wait list has been completed. Since XXX1.WAV is registered in the transfer completion wait list and transfer of XXX1.JPG with the same filename has been completed, the control unit **101** advances the processing to step **S1109**.

[0248] In step **S1109**, the control unit **101** starts transfer of a file in the transfer completion wait list in the FTP session determined to be available in step **S1106** and advances the processing to step **S1110**. Since XXX1.WAV is registered in the transfer completion wait list, the control unit **101** starts transfer of XXX1.WAV in FTP session 1.

[0249] In step **S1110**, the control unit **101** deletes the file for which transfer has been started in step **S1109** from the transfer completion wait list and returns the processing to step **S1106**. The control unit **101** deletes XXX1.WAV for which transfer has been started in step **S1109** from the transfer completion wait list. In this case, a state in which no audio file is registered in the transfer completion wait list is entered.

[0250] In step **S1106**, the control unit **101** determines whether an FTP session is available. At this time, XXX1.WAV is being transferred in FTP session 1 and XXX2.JPG is being transferred in FTP session 2, and thus, the control unit **101** waits until it is determined that an FTP session is available. Then, the control unit **101** determines that transfer of XXX2.JPG in FTP session 2 has been completed and FTP session 2 has become available and advances the processing to step **S1107**.

[0251] In step **S1107**, the control unit **101** determines that no audio file is registered in the transfer completion wait list and advances the processing to step **S1111**.

[0252] In step **S1111**, the control unit **101** starts transfer of the image file in the FTP session determined to be available in step **S1106** and advances the processing to step **S1113**. As illustrated in FIG. **10**, the control unit **101** starts transfer of XXX3.CR3 in FTP session 2.

[0253] In steps **S1113** and **S1114**, the control unit **101** deletes XXX3.CR3 from the transfer list and, since files XXX3.JPG to XXX4.WAV remain in the transfer list, determines that a file remains in the transfer list and returns the processing to step **S1104**.

[0254] This concludes the processing for transferring XXX3.CR3.

[0255] In steps **S1104** and **S1105**, the control unit **101** determines that the first file in the transfer list is XXX3.JPG and XXX3.JPG is an image file and thus advances the processing to step **S1106**.

[0256] In step **S1106**, the control unit **101** determines whether an FTP session is available. When XXX3.JPG is referenced in step **S1104**, XXX1.WAV is being transferred in FTP session 1 and XXX3.CR3 is being transferred in FTP session 2, and thus, the control unit **101** waits until it is determined that an FTP session is available. Then, the control unit **101** determines that transfer of XXX1.WAV in FTP session 1 has been completed and FTP session 1 has become available and advances the processing to step **S1107**.

[0257] In step **S1107**, the control unit **101** determines whether an audio file is registered in the transfer completion wait list. In this case, since no audio file is registered in the transfer completion wait list, the control unit **101** advances the processing to step **S1111**.

[0258] In step **S1111**, the control unit **101** starts transfer of the image file in the FTP session determined to be available in step **S1106** and advances the processing to step **S1113**. As illustrated in FIG. **10**, the control unit **101** starts transfer of XXX3.JPG in FTP session 1.

[0259] In steps **S1113** and **S1114**, the control unit **101** deletes XXX3.JPG from the transfer list and, since files XXX3.WAV to XXX4.WAV remain in the transfer list, determines that a file remains in the transfer list and returns the processing to step **S1104**.

[0260] This concludes the processing for transferring XXX3.JPG.

[0261] In steps **S1104** and **S1105**, the control unit **101** determines that the first file in the transfer list is XXX3.WAV and XXX3.WAV is an image file and thus advances the processing to step **S1112**.

[0262] In step **S1112**, the control unit **101** registers the audio file in the transfer completion wait list and advances the processing to step **S1113**. The control unit **101** registers XXX3.WAV in the transfer completion wait list.

[0263] In steps **S1113** and **S1114**, the control unit **101** deletes XXX3.WAV from the transfer list and, since files XXX4.JPG and XXX4.WAV remain in the transfer list, determines that a file remains in the transfer list and returns the processing to step **S1104**.

[0264] This concludes the processing for transferring XXX3.WAV.

[0265] In steps **S1104** and **S1105**, the control unit **101** determines that the first file in the transfer list is XXX4.JPG and XXX4.JPG is an image file and thus advances the processing to step **S1106**.

[0266] In step **S1106**, the control unit **101** determines whether an FTP session is available. When XXX4.JPG is referenced in step **S1104**, XXX3.JPG is being transferred in FTP session 1 and XXX3.CR3 is being transferred in FTP session 2, and thus, the control unit **101** waits until it is determined that an FTP session is available. Then, the control unit **101** determines that transfer of XXX3.CR3 in FTP session 2 has been completed and FTP session 2 has become available and advances the processing to step **S1107**.

[0267] In step **S1107**, the control unit **101** determines whether an audio file is registered in the transfer completion wait list. In this case, since an audio file XXX3.WAV is registered in the transfer completion wait list, the control unit **101** advances the processing to step **S1108**.

[0268] In step **S1108**, the control unit **101** determines whether transfer of all image files with the same file number as the audio file in the transfer completion wait list has been completed. Since XXX3.WAV is registered in the transfer completion wait list and transfer of XXX3.CR3, which is a file with the same filename, has been completed but transfer of XXX3.JPG has not been completed, the control unit **101** advances the processing to step **S1111**.

[0269] In step **S1111**, the control unit **101** starts transfer of the image file in the FTP session determined to be available in step **S1106** and advances the processing to step **S1113**. As illustrated in FIG. **10**, the control unit **101** starts transfer of XXX4.JPG in FTP session 2.

[0270] In steps S**1113** and S**1114**, the control unit **101** deletes XXX4.JPG from the transfer list and, since XXX4.WAV remains in the transfer list, determines that a file remains in the transfer list and returns the processing to step S**1104**.

[0271] This concludes the processing for transferring XXX4.JPG.

[0272] In steps S**1104** and S**1105**, the control unit **101** determines that the first file in the transfer list is XXX4.WAV and XXX4.WAV is an audio file and thus advances the processing to step S**1112**.

[0273] In step S**1112**, the control unit **101** registers the audio file in the transfer completion wait list and advances the processing to step S**1113**. In this case, the control unit **101** registers XXX4.WAV in the transfer completion wait list, and a state in which XXX3.WAV and XXX4.WAV are registered in the transfer completion wait list is entered.

[0274] In steps S**1113** and S**1114**, the control unit **101** deletes XXX4.WAV from the transfer list and, since no file remains in the transfer list, advances the processing to step S**1115**.

[0275] This concludes the processing for transferring XXX4.WAV.

[0276] In step S**1115**, the control unit **101** determines whether a file remains in the transfer completion wait list. When it is determined that a file remains in the transfer completion wait list, the control unit **101** advances the processing to step S**1116**, and when it is determined that no file remains in the transfer completion wait list, the control unit **101** ends the processing. In this case, since XXX3.WAV and XXX4.WAV remain in the transfer completion wait list, the control unit **101** advances the processing to step S**1116**.

[0277] In step S**1116**, the control unit **101** determines whether there is an available FTP session. The control unit **101** waits until it is determined that there is an available FTP session, and when it is determined that there is an available FTP session, the control unit **101** advances the processing to step S**1117**. At this time, XXX3.JPG is being transferred in FTP session 1 and XXX4.JPG is being transferred in FTP session 2, and thus, the control unit **101** waits until it is determined that an FTP session is available. Then, the control unit **101** determines that transfer of XXX3.JPG in FTP session 1 has been completed and FTP session 1 has become available and advances the processing to step S**1117**.

[0278] In step S**1117**, the control unit **101** determines whether transfer of all image files with the same file name as the audio file in the transfer completion wait list has been completed. When it is determined that all have been completed, the control unit **101** advances the processing to step S**1118** and when it is determined that all have not been completed, the control unit **101** returns the processing to step S**1116**. In this case, regarding XXX3.WAV and XXX4.WAV in the transfer completion wait list, since transfer of XXX3.JPG with the same file name as XXX3.WAV has been completed and transfer of all files with the same file name has been completed, the control unit **101** advances the processing to step S**1118**.

[0279] In step S**1118**, the control unit **101** starts transfer of an audio file in the transfer completion wait list in the FTP session determined to be available in step S**1116** and advances the processing to step S**1119**. The control unit **101** starts transfer of XXX3.WAV in FTP session 1.

[0280] In step S**1119**, the control unit **101** deletes the audio file for which transfer has been started in step S**1118** from the transfer completion wait list and returns the processing to step S**1115**. In this case, the state is such that XXX4.WAV remains in the transfer completion wait list.

[0281] In step S**1115**, since XXX4.WAV remains in the transfer completion wait list, the control unit **101** performs processing from step S**1116** to step S**1119**. When transfer of XXX4.JPG with the same file name as XXX4.WAV is completed, the control unit **101** starts transfer of XXX4.WAV in FTP session 2 in which XXX4.JPG has been transferred, deletes XXX4.WAV from the transfer completion wait list, and returns the processing to step S**1115**. At this time, a state in which there is no file in the transfer completion wait list is entered.

[0282] In step S**1115**, the control unit **101** determines there is no audio file in the transfer completion wait list and ends the processing.

[0283] As described above, according to Embodiment 3, when transferring an image file and an audio file attached to the image file in a plurality of FTP sessions, when there are two files with the same file name including the audio file, the audio file is transferred last in the same FTP session, and when there are three files with the same file name including the audio file, RAW and JPEG/HEIF are transferred in available FTP sessions in that order and the audio file is transferred in the same FTP session as the file for which transfer has been completed last among RAW and JPEG/HEIF. This makes it possible to perform control such that when transferring an image file and an audio file attached to the image file in a plurality of FTP sessions, the audio file arrives at the FTP server later than the image file.

[0284] In the present embodiment, processing in which RAW and JPEG/HEIF are transferred in available FTP sessions in that order and the audio file is transferred in the same FTP session as the file for which transfer has been completed last among RAW and JPEG/HEIF has been described, but a configuration may be taken so as to transfer a plurality of image files in available FTP sessions in order from the file with the largest size and transfer the audio file in the same FTP session as the file that is an image file with the same file name and for which transfer has been completed last. Thus, for example, even where files with the same file name are a combination of a plurality of types of files, such as JPEG, HEIF, and audio file, by performing transfer in an available FTP session in order from the image file with the largest size and transferring the audio file in the same FTP session as the file for which transfer has been completed last among files with the same file name, it is possible to perform control such that the audio file arrives at the FTP server later than the image file.

[0285] Further, a configuration may be taken so as not to consider the order in which image files are transferred and transfer the audio file in the same FTP session as the image file which is a file with the same file name and for which transfer has been completed last. This eliminates the need to rearrange image files before start of transfer.

[0286] According to the present disclosure, it is possible to perform control such that a voice memo arrives later than an image file when transferring the image file to which the voice memo is attached to an external apparatus.

Other Embodiment

[0287] Embodiment(s) of the present disclosure can also be realized by a computer of a system or apparatus that reads out and executes computer executable instructions (e.g., one or more programs) recorded on a storage medium (which may also be referred to more fully as a 'non-transitory computer-readable storage medium') to perform the functions of one or more of the above-described embodiment(s) and/or that includes one or more circuits (e.g., application specific integrated circuit (ASIC)) for performing the functions of one or more of the above-described embodiment(s), and by a method performed by the computer of the system or apparatus by, for example, reading out and executing the computer executable instructions from the storage medium to perform the functions of one or more of the above-described embodiment(s) and/or controlling the one or more circuits to perform the functions of one or more of the above-described embodiment(s). The computer may comprise one or more processors (e.g., central processing unit (CPU), micro processing unit (MPU)) and may include a network of separate computers or separate processors to read out and execute the computer executable instructions. The computer executable instructions may be provided to the computer, for example, from a network or the storage medium. The storage medium may include, for example, one or more of a hard disk, a random-access memory (RAM), a read only memory (ROM), a storage of distributed computing systems, an optical disk (such as a compact disc (CD), digital versatile disc (DVD), or Blu-ray Disc (BD)™), a flash memory device, a memory card, and the like.

[0288] While the present disclosure has been described with reference to exemplary embodiments, it is to be understood that the present disclosure is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

1. A communication apparatus comprising:

a connection unit that connects with an external apparatus so as to be capable of communication; and

a control unit that establishes a plurality of data transfer paths with the external apparatus with which a connection has been established by the connection unit, selects a data transfer path on which transfer is performed, for each image file from among the plurality of data transfer paths, and transfers the image file,

wherein the control unit

selects a data transfer path on which transfer of a file is possible from among the plurality of data transfer paths, and

in a case of transferring an audio file attached to the image file, selects a data transfer path on which transfer of the image file to which the audio file is attached has been performed.

2. The communication apparatus according to claim **1**, wherein the control unit

determines whether an audio file is attached to an image file to be transferred to the external apparatus, and

in a case of transferring an image file to which the audio file is attached, selects a data transfer path on which transfer of a file is possible from among the plurality of data transfer paths and transfers the image file.

3. The communication apparatus according to claim **1**, further comprising:

a selection unit that selects an image file to be transferred to the external apparatus from among a plurality of image files.

4. The communication apparatus according to claim **3**, wherein the control unit

rearranges the image file selected by the selection unit and an audio file attached to the image file in order of transfer to the external apparatus and registers the image file and the audio file in a transfer list, and

in a case where transfer of a file registered in the transfer list has been completed, deletes the file for which transfer has been completed from the transfer list.

5. The communication apparatus according to claim **4**, wherein

the control unit performs rearrangement such that the audio file attached to the image file is transferred later than the image file to which the audio file is attached.

6. The communication apparatus according to claim **1**, wherein the control unit

in a case where after transferring the image file to which the audio file is attached, there is no data transfer path on which transfer of a file is possible among the plurality of data transfer paths, registers in a transfer pending list a file to be transferred next to the external apparatus on the same data transfer path, and

in a case where transfer of a file registered in the transfer pending list has been completed, deletes the file for which transfer has been completed from the transfer pending list.

7. The communication apparatus according to claim **6**, wherein the control unit

registers in the transfer pending list an audio file to be transferred next to the external apparatus on the same data transfer path

in a case where transfer of a file has become possible on the same data transfer path, transfer the audio file registered in the transfer pending list.

8. The communication apparatus according to claim **4**, wherein

the plurality of image files include image files in different file formats, and

the control unit transfers image files in different file formats to which the audio file is attached on the same data transfer path.

9. The communication apparatus according to claim **8**, wherein

after transferring the image files in different file formats, the control unit transfers the audio file.

10. The communication apparatus according to claim **9**, wherein

the different file formats include at least JPEG and RAW, and

after transferring JPEG and RAW to the external apparatus in that order, the control unit transfer the audio file.

11. The communication apparatus according to claim **8**, wherein

the control unit transfers the audio file on the same data transfer path as an image file for which completion of transfer is last among the plurality of image files in different file formats to which the audio file is attached.

12. The communication apparatus according to claim **11**, wherein

the different file formats include at least JPEG and RAW, and

the control unit transfers the audio file on the same data transfer path as an image file for which completion of transfer is last among JPEG and RAW.

13. The communication apparatus according to claim **1**, wherein

the control unit transfers a plurality of image files in different sizes to which the audio file is attached in order of the sizes and transfers the audio file on the same data transfer path as an image file for which transfer has been completed last.

14. A control method of a communication apparatus having a connection unit that connects with an external apparatus so as to be capable of communication, the method comprising:

performing control so as to establish a plurality of data transfer paths with the external apparatus with which a connection has been established by the connection unit, select a data transfer path on which transfer is performed, for each image file from among the plurality of data transfer paths, and transfer the image file,

wherein in the control,

a data transfer path on which transfer of a file is possible is selected from among the plurality of data transfer paths, and

in a case of transferring an audio file attached to the image file, a data transfer path on which transfer of the image file to which the audio file is attached has been performed is selected.

15. A non-transitory computer-readable storage medium storing a program for causing a computer to function as a communication apparatus comprising:

a connection unit that connects with an external apparatus so as to be capable of communication; and

a control unit that establishes a plurality of data transfer paths with the external apparatus with which a connection has been established by the connection unit, selects a data transfer path on which transfer is performed, for each image file from among the plurality of data transfer paths, and transfers the image file,

wherein the control unit

selects a data transfer path on which transfer of a file is possible from among the plurality of data transfer paths, and

in a case of transferring an audio file attached to the image file, selects a data transfer path on which transfer of the image file to which the audio file is attached has been performed.

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